

**VIETNAM NATIONAL UNIVERSITY, HANOI
INTERNATIONAL SCHOOL**

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PROJECT REPORT
MACHINE LEARNING & DEEP LEARNING

**Develop a Machine Learning and
Deep Learning System
to Detect Customer Emotions from Comments**

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Abstract

In the context of the ever-increasing number of online comments and reviews, automated analysis of customer sentiment plays a crucial role in supporting businesses in evaluating product and service quality and enhancing customer experience. This report presents the development process of a Machine Learning and Deep Learning system to detect customer sentiment from Vietnamese text comments.

The research uses a dataset of customer comments collected and pre-processed through data cleaning, text normalization, and feature encoding. Multiple machine learning and deep learning models were built and evaluated, including traditional classification models and modern linguistic models based on the Transformer architecture such as PhoBERT. Data was divided into training, validation, and testing sets to ensure objectivity in the evaluation process.

The performance of the models was evaluated using the metrics Accuracy, Precision, Recall, and F1-score. Experimental results show that deep learning models, especially PhoBERT, are better able to exploit the semantics and context of the Vietnamese language, thereby achieving higher sentiment classification efficiency compared to traditional methods. The research confirms the feasibility of applying machine learning and deep learning techniques in customer sentiment analysis, contributing to supporting businesses in effectively exploiting and analyzing user feedback.

1. Introduction

In recent years, the rapid development of e-commerce platforms, social networks, and online review systems has generated a large amount of customer feedback data. These comments and reviews contain crucial information about user emotions, satisfaction levels, and experiences with products and services. Automated analysis of customer sentiment not only helps businesses better understand customer needs but also supports decision-making, improves service quality, and builds more effective business strategies.

Sentiment detection from text is a key problem in the field of Natural Language Processing (NLP). Many previous studies have applied machine learning methods such as Logistic Regression, Naive Bayes, and Random Forest to solve the sentiment classification problem. These methods have the advantages of being easy to implement, having a fast training time, and requiring low computational resources. However, the ability to extract semantic and contextual information from texts is limited, especially for Vietnamese a language with many complex grammatical and semantic features.

To improve classification efficiency, deep learning methods such as Recurrent Neural Networks (RNNs) and Convolutional Neural Networks (CNNs) have been widely studied and applied. These models are capable of automatically learning features from data and achieving better results on various sentiment datasets. Recently, pre-trained language models based on the Transformer architecture, notably PhoBERT for Vietnamese, have shown superior performance due to their ability to understand context more deeply and represent language more effectively.

Although there have been many studies on sentiment analysis, comprehensive evaluation and comparison between traditional machine learning methods, deep learning, and pre-trained language models on the same Vietnamese dataset remain limited. Furthermore, many studies focus only on the accuracy of the models without fully considering the differences in generalizability, complexity, and practical applicability of each method.

Based on these issues, this report focuses on building and evaluating a system for detecting customer sentiment from Vietnamese text comments. The team trained and compared six models, including Logistic Regression, Naive Bayes, Random Forest, RNN, CNN, and Teacher PhoBERT, on the same dataset. The models were evaluated using Accuracy, Precision, Recall, and F1-score metrics to determine the most suitable method for the customer sentiment analysis problem.

2. Materials and Methods

2.1. Dataset and Materials

a. Data Sources

In this project, the team uses the Vietnamese Sentiment Analyst dataset obtained from the Kaggle platform. This dataset consists of Vietnamese customer comments, suitable for the problem of analyzing sentiment in text. Using a publicly available dataset from Kaggle provides the team

with a readily available, clear, and convenient data source for training and evaluating machine learning and deep learning models.

The dataset used in this project aims to build a system for detecting customer sentiment from comments. Each data sample includes comment content, sentiment labels, and customer ratings. The main goal of the dataset is to help the model learn how to classify customer sentiment based on text content.

Dataset source: [Kanggle Dataset](#)

b. Dataset Size

The dataset contains a total of 31,460 samples, divided into three emotion categories: positive, negative, and neutral. Specifically:

- The positive category has 20,093 samples (63.9%).
- The negative category has 6,669 samples (21.2%).
- The neutral category has 4,698 samples (14.9%).

=> It can be seen that the number of positive comments is the highest, while the number of neutral comments is smaller. Therefore, the dataset shows a slight imbalance between the categories.

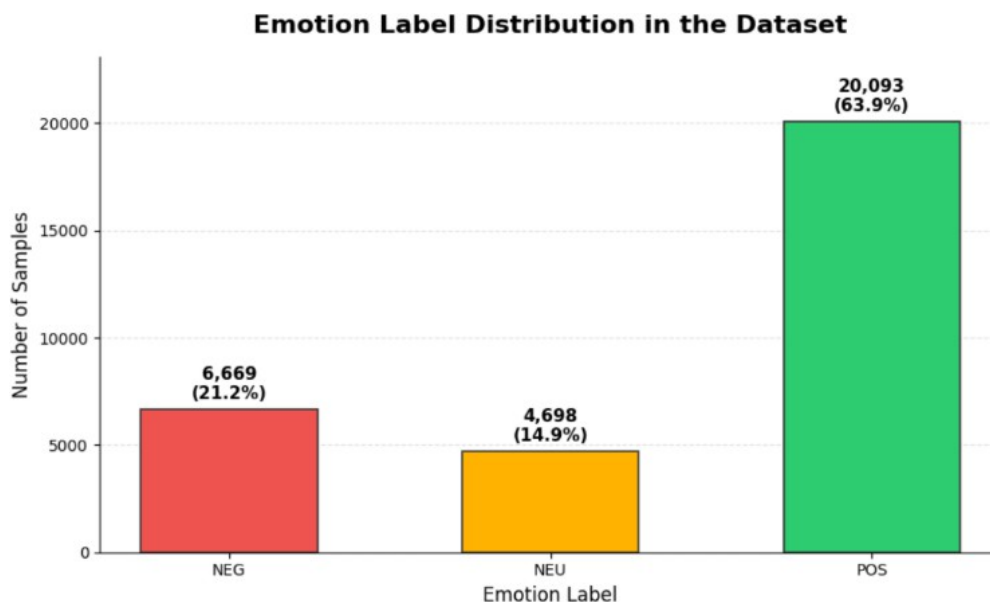


Figure 1: Emotion Label Distribution in the Dataset.

c. Data partitioning and test set

To ensure objectivity in model training and evaluation, the dataset is divided into two sets:

Table 2: Summary of the dataset used in this study.

Subset	Number of Images/Samples	Number of Labels/Instances	Percentage
Training	25.168	25.168	80%
Testing	6292	6292	20%

Overall, the dataset is suitable for the customer sentiment classification problem because it contains complete comment content and corresponding sentiment labels. However, since the sample size between classes is not perfectly balanced, in addition to accuracy, the team also used metrics such as Precision, Recall, and F1-score to evaluate the model more objectively.

A		B		C
Comment		Label		Rate
1 Áo bao đẹp a!!		POS		5
2 Tuyệt vời !		POS		5
3 2day ao không giống trong.		NEG		1
4 Mùi thơm,bôi lên da mềm da.		POS		5
5 Vải đẹp, dày dặn.		POS		5
6 Hàng rất đẹp, rất chi là ưng ý.		POS		5
7 Chất lượng sản phẩm tốt, date dài.		POS		5
8 Ăn nói và thái độ phục vụ tốt.		POS		4
9 Đóng gói sản phẩm chắc chắn.		POS		5
10 tất sơn hết ca chưa đúng mà vay r.		NEG		2
11 Shop phục vụ rất tốt.		POS		5
12 Mặc thì cũng đc.		NEU		3
13 Chất vải khỏi chê.		POS		5
14 Thời gian giao hàng rất nhanh.		POS		5
15 Chất lượng sản phẩm tuyệt vời.		POS		4
16 vải hơi thô cứng Thời gian giao hàng nhanh.		POS		5
17 Chất lượng sp chưa thật sự đẹp nhe shop!		POS		4
18 Rất đáng tiền Thời gian giao hàng rất nhanh.		POS		5
19 Quần rất đẹp mặc vừa vặn.		POS		5
20 Cảm giác mua hàng bị hớ thật tệ!		NEG		2
21 Khi mua về nên đi sửa lại.		NEG		2
22 Với giá này thì sản phẩm tạm ổn chưa đc gọi là đẹp lắm.		NEU		3
23 Rất đáng tiền Thời gian giao hàng rất nhanh Chất lượng sản phẩm tuyệt vời.		POS		5
24 Giá cả chấp nhận được.		POS		4
25 Nhung là rất ổn ♥️.		POS		5
26 Áo quá đẹp luôn nếu không muốn nói là đẹp, may quá có áo mới đi làm cty mới, shop còn mẫu nào trắng nữa để mình mua.		POS		5

Figure 2: Representative examples from the dataset.

2.2. Data Preprocessing

Before feeding data into the models, the team performed a preprocessing step to clean the text data and format it appropriately for training. Since the input data consisted of Vietnamese customer comments, the preprocessing focused primarily on normalizing the text content.

First, the team checked for missing values in the dataset to ensure that important columns such as comment content, sentiment labels, and ratings were not empty. Then, comments were converted to lowercase to reduce the difference between identical words with different capitalization. Next, the team removed unnecessary characters such as special characters, extra spaces, and some punctuation marks that did not significantly affect the meaning of the sentence.

After the data cleaning step, sentiment labels were converted to numerical form so the model could process them. Specifically, sentiment labels such as positive, negative, and neutral were encoded into their corresponding values. For traditional machine learning models like Logistic Regression, Naive Bayes, and Random Forest, text data is converted into feature vectors using text representation methods. For deep learning models like RNN and CNN, text data is tokenized, then converted into numerical sequences and normalized in length for input into the

model.

The pre-processed dataset is divided into two parts: a training set and a test set, in an 80% - 20% ratio. The training set is used for the model to learn features from the data, while the test set is used to evaluate the model's predictive ability on the new data. Data preprocessing helps reduce noise in the text, minimize errors during training, and contribute to improving the model's sentiment classification performance.

The data preprocessing process is performed based on common steps in natural language processing, including text cleaning, label encoding, splitting training/test data, and converting text into numerical form. In this approach, data splitting is performed using `train_test_split`, text is represented using TF-IDF for traditional machine learning models, while deep learning models use tokenization to convert text into numerical sequences. [7], [8]

2.3. Model Architecture or Proposed Method

2.3.1. Deep Learning Modeling Approaches

In this study, three deep learning architectures were selected and evaluated in parallel for the task of detecting customer sentiment from Vietnamese comments: RNN, CNN, and Teacher PhoBERT. Because the input data consists of short comments collected from social media environments with inconsistent text quality, containing many abbreviations, slang, and diverse expressions of emotion, parallel testing of multiple architectures helps determine the model complexity required to effectively capture sentiment signals in Vietnamese text. It also provides a basis for comprehensive comparison between inference speed, accuracy, and practical feasibility.

a, RNN (Recurrent Neural Network)

In terms of architecture, the model is built on two Bidirectional LSTM layers with a hidden size of 128, combined with an Embedding layer of $\text{vocab} \times 128$ at the input to map each token to a dense vector space. A Dropout layer is applied after the feedback layers to control overfitting, followed by a Dense layer that fully connects with the number of outputs equal to the number of emotion layers and a Softmax activation function to create a probability distribution on the categorical labels.

Regarding its role in the topic, emotions in Vietnamese are often strongly dependent on context in both directions of the sentence. A word with a negative connotation like “terrible” can completely transform into a positive meaning in the phrase “not bad at all” when read in combination with surrounding words. Traditional one-dimensional LSTMs only process sequences from left to right, thus failing to fully capture contextual relationships. Bi-LSTMs, however, process both directions simultaneously, allowing each token position to be represented with complete contextual information from the entire sentence. Stacking two LSTM layers allows the upper layer to learn higher-level abstract features from the semantic sequence built by the lower layer, increasing the hierarchical representation of the model. In the context of customer sentiment detection, this model serves as a lightweight sequential baseline, suitable for

deployment scenarios requiring a balance between performance and computational resources.

b, CNN (Convolutional Neural Network)

Architecturally, the model applies parallel Conv1D layers with three kernel sizes: 2, 3, and 4, each using 128 independent filters. After each convolutional layer, Global Max Pooling is applied to extract the most prominent features from each filter. Feature vectors from the three kernel branches are then joined along the feature direction to form a composite representation before being passed through the Softmax classification layer.

Regarding its role in the topic, emotional information in customer comments is often encoded at various phrase lengths. Kernel size 2 captures highly polarized emotional bigrams such as “very good” or “terrible”; kernel size 3 captures trigrams with more complex nuances such as “not very good” or “quite satisfied”; while kernel size 4 targets longer phrases with broader emotional contextual meaning. The simultaneous use of three kernel sizes via a multi-scale feature fusion mechanism allows the model to learn emotional features at various levels of linguistic detail, instead of being limited to a single n-gram type. Global Max Pooling ensures positional invariance, meaning the model recognizes emotional phrases regardless of their position in the sentence – a crucial characteristic because customers can place emotional keywords anywhere in a comment. Compared to feedback architectures, CNNs offer significantly faster inference speeds thanks to GPU parallelization capabilities, making them suitable for systems needing to process large volumes of comments in a short amount of time.

c. Teacher PhoBERT

Architecturally, PhoBERT-based consists of 12 Transformer layers with a hidden size of 768 and 12 attention heads, using a BPE tokenizer with a maximum string length of 128 tokens. A pooling layer is applied to contextual representations to create sentence-level representation vectors, followed by a classification layer trained with a Weighted CrossEntropy loss function combined with class weights to handle class imbalances in the training data.

Regarding its role in the project, PhoBERT was chosen as a teacher model representing the upper bound of accuracy in the deep learning group. Unlike general multilingual models like mBERT or XLM-R, PhoBERT was pre-trained specifically on a large-scale Vietnamese corpus including the entire Vietnamese Wikipedia and hundreds of millions of words from news data, giving the model deep understanding of the morphological structure, syntax, and semantics of the Vietnamese language. Tokenizer BPE handles out-of-dictionary terms by breaking them down into subwords, which is particularly important for social media comments that contain many slang words and abbreviations not found in standard dictionaries. Transformer’s multi-headed Self-Attention mechanism allows the model to learn complex semantic relationships between tokens at long distances within sentences, outperforming LSTM and CNN in capturing long-range dependencies. The use of Weighted CrossEntropy Loss, with class weights inversely proportional to frequency of occurrence, directly addresses the class imbalance problem in customer sentiment data, preventing the model from biased towards predicting the majority class and significantly improving the F1-score on the minority class.

2.3.2. *Machine Learning Modeling Approaches*

To provide an objective comparative perspective with deep learning models, the study incorporates three classical machine learning algorithms - Naive Bayes, Logistic Regression, and Random Forest - into the experimental system. Unlike deep learning, which can self-learn features from raw text through end-to-end training, classical algorithms require a distinct manual feature design phase. In this pipeline, features are built using a Dual TF-IDF mechanism combining Word TF-IDF and Char TF-IDF via `FeatureUnion`, creating a representation space that is both rich in vocabulary and robust to the specific linguistic noise of Vietnamese comments. Word TF-IDF, with `max_features = 30,000` and `min_df = 2`, represents text at the word and phrase level, while Char TF-IDF, with `max_features = 50,000`, operates at the character level to capture abbreviation patterns and spelling variations that Word TF-IDF misses. The parameter `sublinear_tf = True` is applied to both to reduce the frequency on a logarithmic scale, lessening the influence of words that appear too frequently but carry little information distinguishing emotional classes.

a. Naive Bayes

In terms of architectural features, the model uses a `ComplementNB` variant with a Laplace smoothing α of 0.05, automatically searched through `GridSearchCV` with a three-class cross-validation test. Predictions are made using `argmax` on log-probability instead of pure probability to avoid arithmetic underflow when calculating in high-dimensional feature spaces.

Regarding its role in the project, `ComplementNB` was chosen over the traditional `MultinomialNB` for a reason with a clear theoretical basis: `ComplementNB` estimates posterior probability by learning from all additional classes instead of from the target class itself, making it significantly more stable when class distribution is uneven - a common characteristic in customer sentiment data where the `NEGATIVE` label often accounts for a lower proportion. With low computational complexity and extremely fast training time, `ComplementNB` serves as the fastest baseline in the entire system, providing a minimum reference point for evaluating the improvement of more complex models.

b. Logistic Regression

In terms of architectural features, the model applies the `OneVsRest` strategy to extend the binary classification problem to multi-class, training an independent binary classifier with sigmoid output for each emotion label. A `liblinear` solver is specified with `max_iter = 1,000`, and the regularization parameter is automatically searched through a three-layer `GridSearchCV`.

Regarding its role in the project, the `OneVsRest` strategy allows each emotion class to learn a separate and independent decision threshold, offering more flexibility compared to the general decision space of `Softmax`, as the boundaries between the `POSITIVE`, `NEGATIVE`, and `NEUTRAL` classes are not symmetrical - a practical characteristic given that the linguistic signals of each emotion in Vietnamese have distinctly different lexical features. The `liblinear` solver was chosen because it is particularly optimized for sparse feature spaces, typical of TF-IDF

vectors where the majority of elements are zero. Beyond classification performance, Logistic Regression also offers advantages in model interpretation - the learned coefficients can be analyzed to identify which words or phrases contribute most to each sentiment label, providing valuable information for understanding the linguistic characteristics of each class in the customer comment dataset.

c. Random Forest

In terms of architectural features, the model is configured with 500 independently trained decision trees using bootstrap sampling. The `max_features = sqrt` parameter limits the number of features considered at each split node, and `class_weight = balanced` automatically adjusts the sample weight inversely proportional to the class frequency. The final prediction is determined through majority voting across all 500 trees.

Regarding its role in the topic, Random Forest introduces nonlinearity into the classical modeling group, allowing for the capture of complex interactions between TF-IDF features that linear Logistic Regression cannot model. The `max_features = sqrt` parameter forces each tree to consider only a random subset of features at each split node, intentionally creating diversity among trees in the ensemble and reducing correlation between them - this is a key mechanism that helps Bagging reduce overall variance without increasing bias. Setting `class_weight = balanced` resolves class imbalance in a similar way to Weighted CrossEntropy Loss in PhoBERT, but implemented at the classical algorithmic level. With 500 trees and a bootstrap aggregation mechanism, Random Forest represents the most powerful configuration in the classical machine learning group, acting as the upper limit of achievable accuracy without the need for deep contextual representations of deep learning, thereby providing a clear quantitative basis for evaluating the improvement that deep learning models bring to the problem of detecting customer sentiment in Vietnamese.

2.3.3. Design Rationale

a, Overall System Architecture

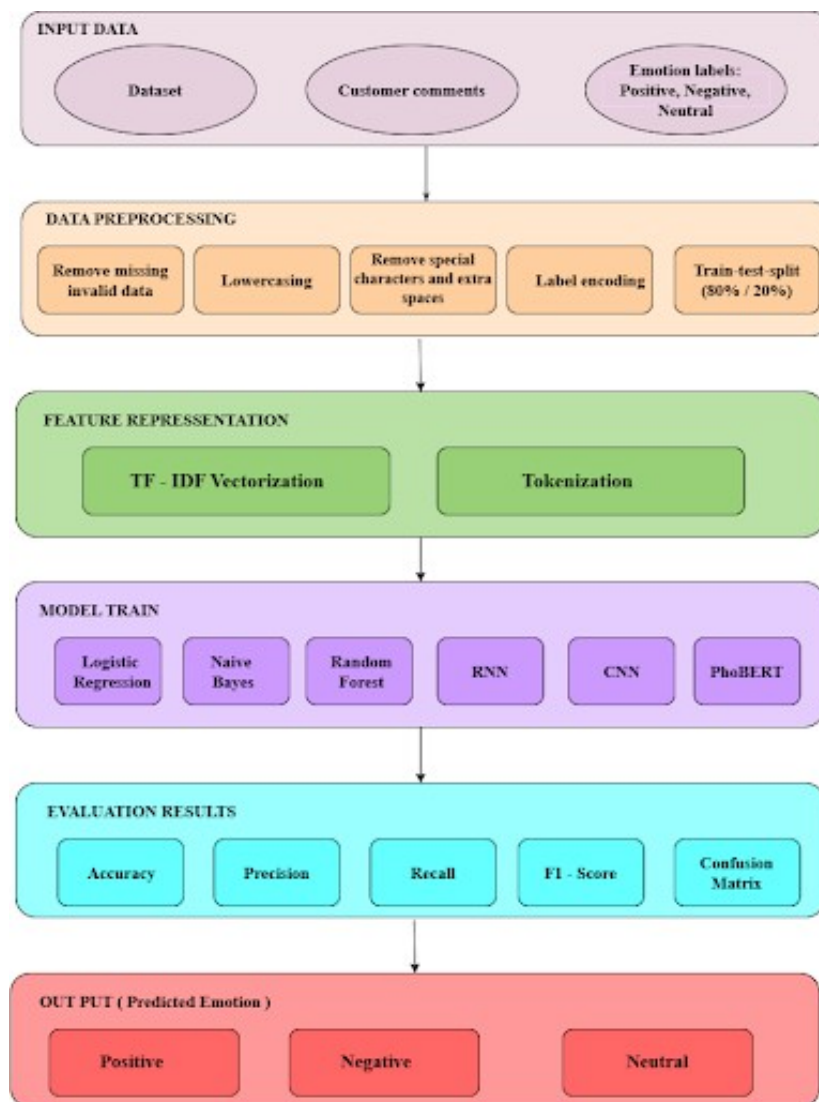


Figure 3: Overall System Architecture .

The image above illustrates the general process of a system for detecting customer sentiment from comments. First, the system uses an input dataset consisting of customer comments and corresponding sentiment labels such as Positive, Negative, and Neutral. Then, the data undergoes preprocessing, including removing missing or invalid data, converting text to lowercase, removing special characters and extra spaces, encoding labels, and finally splitting the data into a training set and a test set in an 80% - 20% ratio.

After the preprocessing step, the data is transformed into suitable features for input into the model. For traditional machine learning models, text is represented using TF-IDF. For deep learning models, the data is processed using tokenization to convert text into digital form. Next, the team trained six models: Logistic Regression, Naive Bayes, Random Forest, RNN, CNN, and PhoBERT.

After training, the models were evaluated using metrics such as Accuracy, Precision, Recall, F1-score, and Confusion Matrix. Finally, the system predicted the emotional response of the comment, categorizing it into one of three classes: Positive, Negative, or Neutral.

b, Data Processing Pipeline Workflow

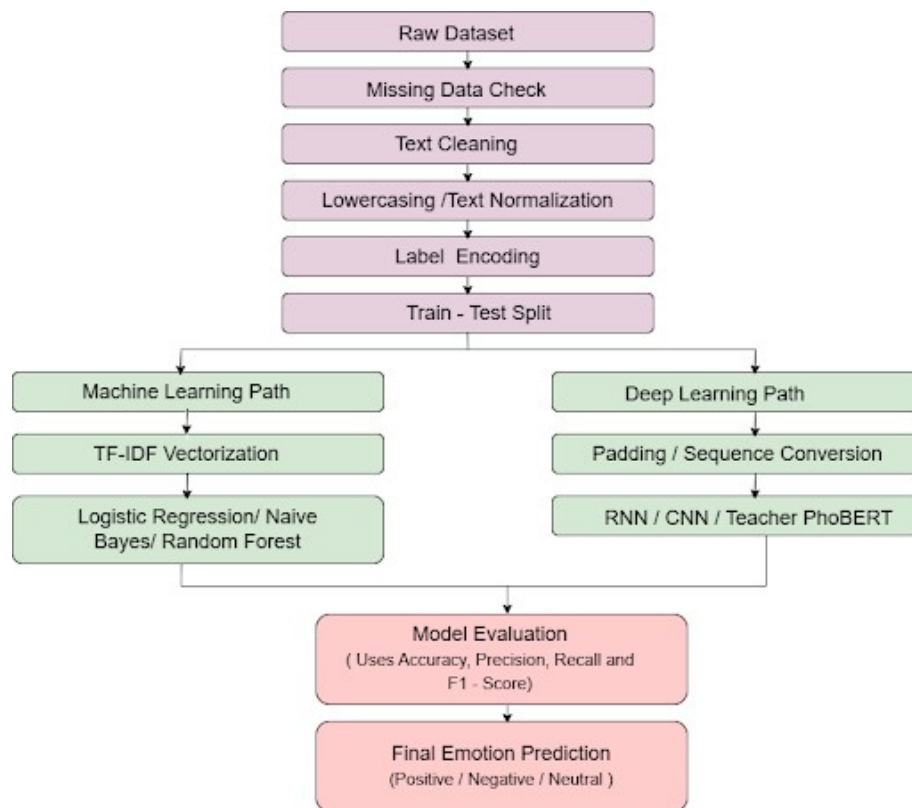


Figure 4: Data Processing Pipeline Workflow.

The image above illustrates the data processing process in a customer sentiment detection system based on comments. First, raw data is fed into the system in the form of customer comments. Then, the data is checked for missing values, followed by text cleaning to remove unnecessary elements. The text is then standardized by converting it to lowercase and applying consistent formatting.

After preprocessing, sentiment labels are encoded to suit the model training process. The dataset is then divided into two parts: a training set and a test set. From here, the data is split into two different processing branches. With traditional machine learning models, text data is converted into feature vectors using TF-IDF and fed into Logistic Regression, Naive Bayes, and Random Forest models. With deep learning models, the data is converted into numerical sequences through padding or sequence conversion, and then fed into RNN, CNN, and Teacher PhoBERT models. Finally, the models are evaluated based on the Accuracy, Precision, Recall, and F1-score metrics. The system's output is the final sentiment label of the comments, comprising three classes: Positive, Negative, and Neutral.

c, Architecture diagram of the deep learning model

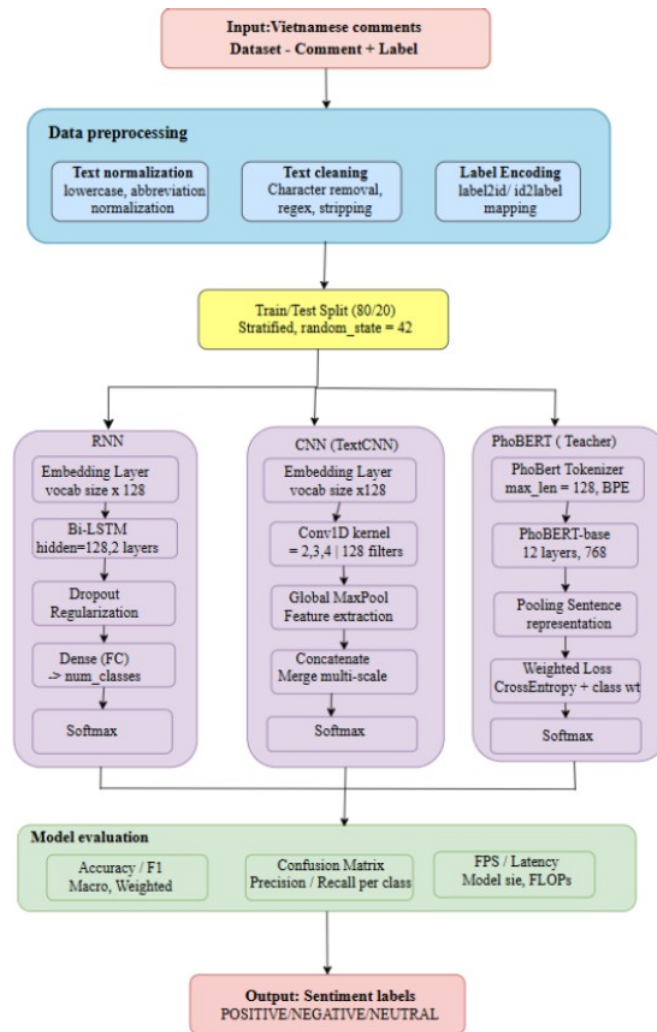


Figure 5: Architecture diagram of the deep learning model .

In the image above, regarding data preprocessing, since the input consists of Vietnamese comments from social media, it contains a lot of noise such as abbreviations, special characters, and non-standard spelling. Therefore, the pipeline is designed in three steps: text normalization, cleaning, and label coding to reduce noise and prepare the data for training. The data is then split at an 80/20 ratio with a stratified strategy to ensure an even distribution of classes between the training and test sets, avoiding bias due to class imbalance. The pipeline uses three parallel models: RNN, CNN, and PhoBERT, to perform multidimensional comparisons between speed and accuracy. The RNN model uses a two-layer Bi-LSTM because Vietnamese emotions are context-dependent in both directions, combined with Dropout to avoid overfitting. The CNN model uses TextCNN with three kernel sizes 2, 3, and 4 in parallel to capture emotional features at various phrase lengths, then Global Max Pooling extracts prominent features regardless of their position in the sentence. The PhoBERT model was chosen because it was pre-trained on a Vietnamese corpus, uses Weighted CrossEntropy Loss to handle class imbalance, and avoids models that are biased towards predicting the majority class. Finally, the evaluation system uses F1 Macro, F1 Weighted, and Confusion Matrix simultaneously instead of Accuracy alone, as

Accuracy is not reliable enough when data is imbalanced. Additionally, FPS and FLOPs metrics help assess feasibility for real-world implementation.

d, Architecture diagram of the machine learning model.

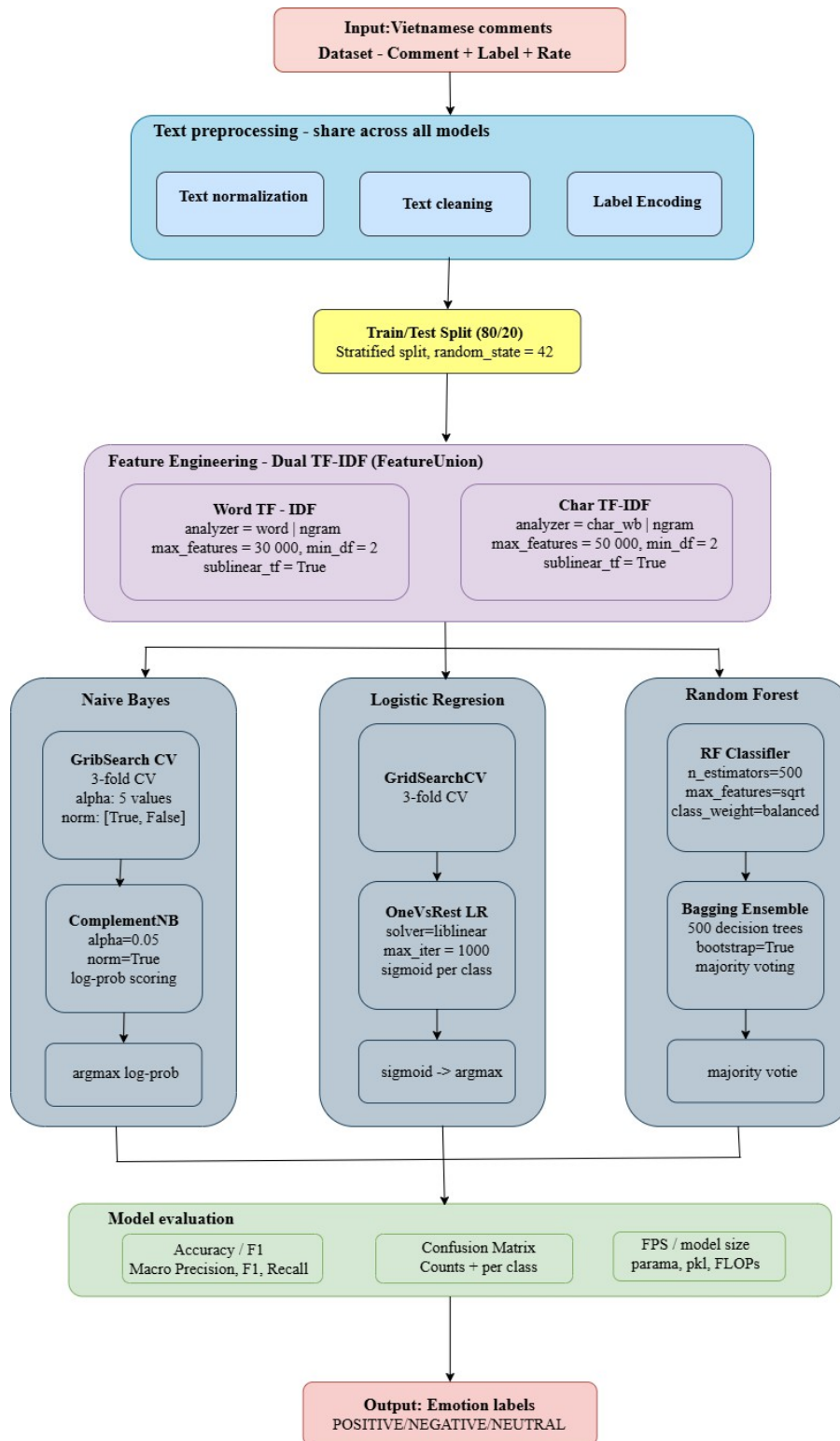


Figure 6: Architecture diagram of the machine learning model .

This pipeline is based on three traditional machine learning algorithms, namely Naive Bayes, Logistic Regression, and Random Forest, which share a common preprocessing stage to ensure a fair comparison. Unlike the deep learning pipeline, the dataset additionally includes a numerical *Rate* feature that provides complementary sentiment information beyond textual content.

To transform text into numerical representations, a `FeatureUnion` combines Word TF-IDF and Character TF-IDF features. Word TF-IDF captures semantic information at the word level, while Character TF-IDF is more effective in handling abbreviations, misspellings, and informal writing commonly found in user comments. This combination produces a richer and more robust feature representation. Furthermore, `sublinear_tf=True` is applied to reduce the influence of excessively frequent terms, whereas `min_df=2` removes extremely rare terms, helping reduce noise and mitigate overfitting.

For model configuration, Complement Naive Bayes is adopted due to its robustness on imbalanced datasets. Logistic Regression employs a One-vs-Rest strategy with the `liblinear` solver, which is well suited for sparse TF-IDF features. Random Forest is configured with 500 decision trees, `class_weight=balanced`, and `max_features=sqrt` to improve class balance handling and model diversity. Hyperparameter optimization for all models is conducted using `GridSearchCV` with 3-fold cross-validation to objectively identify the best-performing configurations.

Finally, model performance is evaluated using F1-score, Confusion Matrix, and computational efficiency metrics, including model size and FLOPs, enabling a comprehensive comparison of both classification effectiveness and deployment feasibility.

2.3.4. Loss Function and Optimization Objective

The optimization objective of the proposed system is to minimize classification errors while improving generalization on unseen customer comments. Different loss functions are employed according to the characteristics of each model.

For deep learning models (RNN, CNN, and PhoBERT), the training objective is to minimize the Cross-Entropy Loss:

$$L = - \sum_{i=1}^N \sum_{c=1}^C y_{i,c} \log(\hat{y}_{i,c}) \quad (1)$$

where $y_{i,c}$ denotes the ground-truth label and $\hat{y}_{i,c}$ represents the predicted probability for class c .

To mitigate the class imbalance problem, Teacher PhoBERT employs Weighted Cross-Entropy Loss:

$$L = - \sum_{i=1}^N \sum_{c=1}^C w_c y_{i,c} \log(\hat{y}_{i,c}) \quad (2)$$

where w_c is the weight assigned to class c . This mechanism increases the contribution of minority classes during training and improves classification balance.

For machine learning models, Logistic Regression minimizes Binary Cross-Entropy Loss, whereas Random Forest optimizes the Gini Impurity criterion:

$$Gini(t) = 1 - \sum_{c=1}^C p(c|t)^2 \quad (3)$$

These optimization strategies enable each model to learn discriminative sentiment features effectively.[1], [3]

2.3.5. Example of Training and Inference Algorithm

The proposed system follows a unified training and inference workflow for all six models. During training, customer comments are preprocessed and transformed into suitable representations. Deep learning models update their parameters through backpropagation and gradient-based optimization according to:

$$\Theta = \Theta - \eta \nabla_{\Theta} L \quad (4)$$

where Θ denotes model parameters, L is the loss function, and η is the learning rate.

For machine learning models, TF-IDF features are extracted from the training data and optimal hyperparameters are selected using GridSearchCV. During inference, unseen comments are processed through the same preprocessing pipeline before being classified into Positive, Negative, or Neutral sentiment categories. The system outputs both the predicted label and its corresponding confidence score.

2.3.6. Detailed Execution and Violation Detection Logic

The implementation process consists of three stages: model optimization, feature extraction, and sentiment classification.

First, all six models are trained using their corresponding optimization strategies. The best configurations are selected according to validation performance. Next, customer comments are normalized and transformed into appropriate feature representations. Deep learning models receive tokenized sequences, while machine learning models use TF-IDF feature vectors.

During prediction, the final sentiment label is determined by selecting the class with the highest probability:

$$\hat{y} = \arg \max_c P(c|x) \quad (5)$$

For Random Forest, the final prediction is obtained through majority voting among all decision trees:

$$\hat{y} = \text{mode}(h_1(x), h_2(x), \dots, h_k(x)) \quad (6)$$

where $h_k(x)$ denotes the prediction of the k -th decision tree.

Finally, the generated predictions are evaluated using Accuracy, Precision, Recall, F1-score, and Confusion Matrix to compare both classification performance and computational efficiency across all models.

2.4. Experimental Setup

All experiments were conducted under a unified setup to ensure a fair comparison between three machine learning models (Naive Bayes, Logistic Regression, Random Forest) and three deep learning models (Bi-LSTM, TextCNN, PhoBERT). The Vietnamese comment dataset was split into 80% training data and 20% testing data using stratified sampling, with `random_state = 42` to ensure reproducibility.

Before training, all comments were preprocessed through text normalization, including lowercasing, removal of special characters, emojis, and extra spaces. For machine learning models, the processed texts were transformed into TF-IDF feature vectors. For deep learning models, texts were tokenized and padded to a fixed length, while PhoBERT utilized the BPE tokenizer with a maximum sequence length of 128 tokens.

The machine learning models were implemented using `scikit-learn` and optimized with `GridSearchCV`. The Bi-LSTM and TextCNN models were trained using the Adam optimizer, whereas PhoBERT was fine-tuned using AdamW with a learning rate scheduler and early stopping strategy. All deep learning experiments were implemented in PyTorch and executed with CUDA acceleration when available.

Model performance was evaluated on the same test set using Accuracy, Precision, Recall, Macro F1-score, Weighted F1-score, and Confusion Matrix. In addition, inference time, model size, and the number of parameters were recorded to assess computational efficiency and deployment feasibility.

Table 3: Main training and implementation settings.

Setting	Value
Programming language	<i>Python</i>
Framework	<i>Scikit-learn / TensorFlow / Keras / PyTorch / Transformers</i>
Input size or feature dimension	<i>TF-IDF up to 80,000 features; sequence length 128 for PhoBERT</i>
Optimizer	<i>Liblinear / ComplementNB / Random Forest / Adam...</i>
Learning rate	<i>0.0005 for RNN/CNN; 2e-5 for PhoBERT</i>
Batch size	<i>64 for RNN/CNN; 16 for PhoBERT</i>
Number of epochs	<i>20 for RNN/CNN; 5 for PhoBERT</i>
Hardware	<i>CPU/GPU runtime environment</i>
Random seed	<i>42</i>

2.5. Evaluation Metrics

To evaluate the performance of the trained models, this study employs several standard evaluation metrics commonly used in text classification tasks, including Accuracy, Precision, Recall, F1-score, Macro Average, Weighted Average, and Confusion Matrix. These metrics are particularly suitable for customer emotion detection because the system classifies each comment into one of three sentiment categories: Positive, Negative, or Neutral.

Accuracy: Accuracy measures the proportion of correctly classified samples among all samples in the test set. It provides an overall indication of model correctness.

$$Accuracy = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Samples}} \quad (7)$$

Precision: Precision measures the proportion of correctly predicted samples of a class among all samples predicted as that class. It indicates the reliability of positive predictions.

$$Precision = \frac{TP}{TP + FP} \quad (8)$$

where TP denotes True Positives and FP denotes False Positives.

Recall: Recall measures the proportion of correctly predicted samples of a class among all actual samples belonging to that class. It reflects the model's ability to identify relevant instances.

$$Recall = \frac{TP}{TP + FN} \quad (9)$$

where FN denotes False Negatives.

F1-score: F1-score is the harmonic mean of Precision and Recall. It is particularly useful when dealing with imbalanced datasets.

$$F1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (10)$$

Macro Average: Macro Average computes the arithmetic mean of a metric across all classes, assigning equal importance to each class regardless of its size.

$$\text{MacroAvg} = \frac{\text{Metric}_{\text{POS}} + \text{Metric}_{\text{NEG}} + \text{Metric}_{\text{NEU}}}{3} \quad (11)$$

Weighted Average: Weighted Average calculates the average metric value across classes while considering the number of samples in each class.

$$\text{WeightedAvg} = \frac{\sum_{i=1}^C (\text{Support}_i \times \text{Metric}_i)}{\sum_{i=1}^C \text{Support}_i} \quad (12)$$

where Support_i represents the number of samples belonging to class i .

Confusion Matrix: A Confusion Matrix is a tabular representation of classification results that summarizes the numbers of correct and incorrect predictions for each class. It helps identify which sentiment categories are frequently confused by the model and provides a detailed understanding of classification behavior.[4], [5]

3. Results

3.1. Quantitative Comparison

Table 4: Performance comparison on the test set.

Model/Method	Main Metric	Precision	Recall	F1-score	Params	FPS
Teacher PhoBERT	81.06	69.33	68.41	68.30	135.00 M	567.18
CNN	79.69	66.38	66.11	65.03	3.95 M	2421.59
Logistic Regression	79.24	66.05	63.37	62.91	0.14 M	10597.38
RNN (Bi-LSTM)	78.16	64.22	63.78	62.89	3.96 M	569.20
Random Forest	77.67	64.71	64.81	64.23	5.06 M	2833.68
Naive Bayes	77.54	64.93	60.90	60.53	0.14 M	11509.44

Table 4 presents the performance comparison of six machine learning and deep learning models evaluated on the Vietnamese customer emotion classification dataset. The results show that Teacher PhoBERT achieved the best overall performance with a Main Metric (Accuracy) of 81.06%, outperforming all remaining models. In addition, Teacher PhoBERT obtained the highest Precision (69.33%), Recall (68.41%), and F1-score (68.30%), demonstrating its strong capability in capturing semantic information from Vietnamese text and correctly classifying

customer emotions. These results confirm the effectiveness of transformer-based language models for sentiment and emotion analysis tasks.

Among the remaining models, CNN achieved the second-best performance with an Accuracy of 79.69%, Precision of 66.38%, Recall of 66.11%, and F1-score of 65.03%. The model demonstrated a good balance between classification performance and computational cost. The convolutional architecture was able to effectively capture local contextual features from customer comments, resulting in better performance than traditional machine learning approaches.

Logistic Regression achieved an Accuracy of 79.24%, which is only slightly lower than CNN. Despite its relatively simple architecture, the model produced competitive results with a Precision of 66.05% and an F1-score of 62.91%. This indicates that traditional machine learning methods can still serve as strong baselines for text classification tasks when appropriate feature representations are used.

The RNN (Bi-LSTM) model obtained an Accuracy of 78.16%, with Precision, Recall, and F1-score values of 64.22%, 63.78%, and 62.89%, respectively. Although recurrent neural networks are designed to capture sequential information in text, the model was unable to surpass CNN or PhoBERT in this experiment. This may be due to the limited ability of RNN-based architectures to model long-range contextual relationships compared with transformer-based models.

Random Forest and Naive Bayes achieved Accuracies of 77.67% and 77.54%, respectively. While both models produced lower overall performance than the deep learning approaches, they still achieved acceptable results for the emotion classification task. Random Forest obtained a slightly higher F1-score (64.23%) than Logistic Regression and RNN, indicating a relatively balanced classification performance. Meanwhile, Naive Bayes delivered the lowest F1-score (60.53%), suggesting limitations in capturing complex semantic patterns within customer comments.

Overall, the experimental results demonstrate that deep learning models generally outperform traditional machine learning methods on the Vietnamese customer emotion classification task. Teacher PhoBERT achieved the highest classification performance due to its powerful contextual language understanding capability. CNN also showed competitive results while requiring significantly fewer parameters than PhoBERT. Although traditional machine learning models such as Logistic Regression, Random Forest, and Naive Bayes offer lower computational complexity, their classification performance remains inferior to the deep learning approaches. Therefore, Teacher PhoBERT can be considered the most effective model in terms of predictive performance, while CNN provides an attractive balance between accuracy and computational efficiency.

3.2. Computational Efficiency

Table 5: Computational complexity comparison.

Model/Method	Parameters	Model Size	FLOPs	FPS / Inference Time
Naive Bayes	<i>0.1364 M</i>	<i>4.15 MB</i>	<i>0.000273 G</i>	<i>11509.44 FPS / 0.0869 ms</i>
Logistic Regression	<i>0.1364 M</i>	<i>2.77 MB</i>	<i>0.000273 G</i>	<i>10597.38 FPS / 0.0944 ms</i>
CNN	<i>3.9470 M</i>	<i>45.21 MB</i>	<i>0.013167 G</i>	<i>2421.59 FPS / 0.4130 ms</i>
RNN (Bi-LSTM)	<i>3.9643 M</i>	<i>45.43 MB</i>	<i>0.007929 G</i>	<i>569.20 FPS / 1.7569 ms</i>
Teacher PhoBERT	<i>135.0006 M</i>	<i>516.95 MB</i>	<i>0.270001 G</i>	<i>567.18 FPS / 1.7631 ms</i>
Random Forest	<i>5.0611 M</i>	<i>424.943 MB</i>	<i>1.208957 G</i>	<i>2833.68 FPS / 2.2204 ms</i>

Besides classification performance, computational efficiency is also an important factor when deploying an emotion detection system in real-world applications. Models with fewer parameters, smaller model sizes, lower computational complexity (FLOPs), and higher inference speed generally require fewer hardware resources and are more suitable for large-scale or real-time deployment scenarios.

To evaluate computational efficiency, the number of parameters, model size, FLOPs, FPS, and inference time were compared, as presented in Table 5. FPS indicates the number of samples that can be processed per second, while inference time represents the average processing time required for a single prediction. In general, higher FPS and lower inference time indicate better computational efficiency.

The results show that Naive Bayes and Logistic Regression are the most computationally efficient models. Both models contain only 0.1364 million parameters and require merely 0.000273 GFLOPs, making them extremely lightweight. Naive Bayes achieved the highest processing speed at 11,509.44 FPS, followed closely by Logistic Regression at 10,597.38 FPS. Their inference times are below 0.1 ms, making them highly suitable for resource-constrained environments and large-scale deployment.

Among the deep learning models, CNN demonstrated the best computational efficiency. With only 3.947 million parameters, a model size of 45.21 MB, and 0.013167 GFLOPs, CNN achieved an impressive speed of 2,421.59 FPS with an inference time of only 0.413 ms. These results indicate that CNN offers an excellent balance between predictive performance and computational cost.

The RNN (Bi-LSTM) model has a similar number of parameters (3.9643 million) and model size (45.43 MB) compared to CNN. However, its inference speed dropped significantly to 569.20 FPS, with an inference time of 1.7569 ms. This reduction can be attributed to the sequential nature of recurrent networks, which limits parallel computation during inference.

Although Teacher PhoBERT achieved the best classification performance, it is also the most resource-demanding deep learning model. The model contains approximately 135 million parameters and occupies 516.95 MB of storage space, which is considerably larger than all

other models. Despite its high computational requirements, Teacher PhoBERT still maintained a relatively fast inference speed of 567.18 FPS. Nevertheless, its large memory footprint may present challenges when deploying the model on devices with limited hardware resources.

Random Forest occupies a unique position in the comparison. While the model achieved competitive classification performance, it required 5.0611 million parameters, a model size of 424.94 MB, and the highest computational complexity among all evaluated models (1.208957 GFLOPs). This indicates that Random Forest is relatively resource-intensive compared with other traditional machine learning methods.

Overall, the computational efficiency analysis reveals a clear trade-off between classification performance and resource consumption. Teacher PhoBERT provides the highest predictive accuracy but requires substantially more storage and computational resources. CNN offers the most balanced solution by combining strong classification performance with high inference speed and moderate model complexity. Meanwhile, Logistic Regression and Naive Bayes are highly attractive for deployment scenarios where computational resources are limited and real-time processing is a priority. Therefore, the final model selection should consider both classification performance and computational efficiency based on the requirements of the target application.

3.3. Qualitative Results

In addition to quantitative metrics, customer emotion detection systems were also evaluated through various real-world comment scenarios to assess their ability to recognize emotions in different contexts. The qualitative analysis results indicate that the models, particularly Teacher PhoBERT and CNN, are capable of effectively handling many common types of comments found on online platforms.

Case 1: Comments with Explicit Emotional Expressions

For comments containing direct emotional expressions such as ‘Very satisfied’, ‘Excellent service’, ‘The product is excellent’, ‘Extremely disappointed’, or ‘Never come back’, the system generally produces accurate predictions with high confidence. The models can easily identify keywords associated with positive or negative sentiment and assign the appropriate emotion labels. While most models perform well in this scenario, Teacher PhoBERT consistently provides more stable and accurate predictions due to its stronger contextual understanding of Vietnamese language.

Case 2: Comments Containing Mixed Emotions

Many customer reviews include both positive and negative aspects within the same comment, such as ‘The product works well, but the delivery was too slow’ or ‘The support staff were helpful, but the product quality was not as good as expected’. These cases are more challenging because the sentiment is not entirely positive or negative. Experimental observations indicate that Teacher PhoBERT and CNN handle such situations more effectively because they can capture relationships between different parts of a sentence. In contrast, traditional machine learning models such as Naive Bayes and Logistic Regression may focus primarily on prominent

keywords, which can lead to incorrect predictions.

Case 3: Comments with Informal Language and Slang

In practice, comments on social media and e-commerce platforms often contain abbreviations, slang, emojis, or non-standard spellings, such as “great product”, “super fast delivery”, “poor service”, or expressions with repeated characters like “sooooo beautiful” and “love ittt”. Such data can make emotion classification more difficult. Nevertheless, deep learning models maintain relatively strong recognition capabilities by learning semantic patterns from the training data. In particular, Teacher PhoBERT demonstrates greater robustness when processing colloquial and non-standardized comments.

Case 4: Comments with Implicit Emotional Meaning

Some comments do not contain explicit emotional words but still convey positive or negative attitudes, such as ‘I will consider it more carefully next time’, ‘It is not the same as advertised’, or ‘It is acceptable’. These cases require the model to understand context and implicit meanings. The observations show that Teacher PhoBERT performs better than the other models by leveraging contextual knowledge acquired during pre-training. However, very short or ambiguous comments may still lead to misclassification.

Case 5: Long Comments with Multiple Contexts

For lengthy comments containing multiple sentences and diverse opinions, the ability to preserve contextual information becomes particularly important. Deep learning models, especially Teacher PhoBERT, demonstrate superior capability in retaining contextual information compared with traditional machine learning models. Consequently, they can provide more accurate predictions when customer emotions are expressed across multiple sentences or paragraphs.

Overall, the qualitative evaluation demonstrates that the proposed system is capable of effectively recognizing customer emotions across a wide range of real-world situations. Deep learning models, particularly Teacher PhoBERT, exhibit clear advantages in semantic understanding, Vietnamese natural language processing, and recognition of subtle emotional cues. These observations are consistent with the quantitative results presented in Table 4, where Teacher PhoBERT achieved the highest classification performance among all evaluated models.

3.3.1. Training results of Deep Learning algorithms

a, RNN

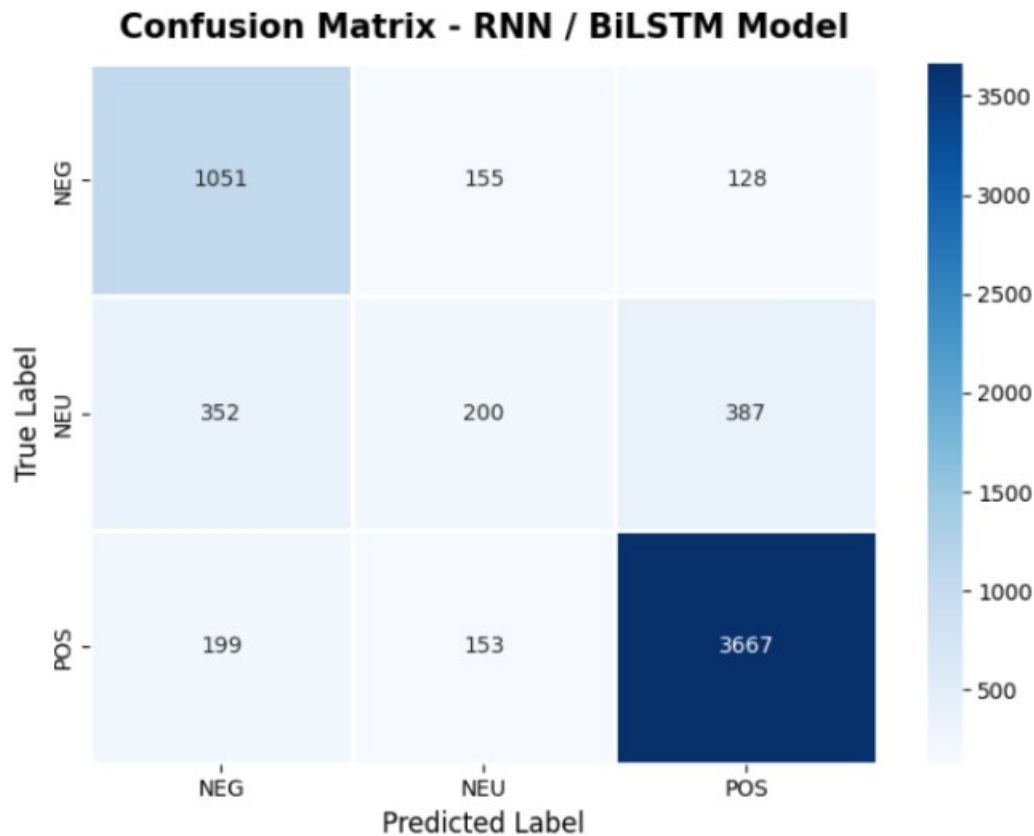


Figure 7: Confusion matrix of the RNN (Bi-LSTM) model .

Figure 7 presents the confusion matrix of the RNN (Bi-LSTM) model on a test set with three emotion classes: Negative (NEG), Neutral (NEU), and Positive (POS). The model achieved an accuracy of 78.16%, Precision of 64.22%, Recall of 63.78%, and an F1-score of 62.89%.

The results show that the model performed best on the Positive (POS) class, with 3,667 samples correctly classified. Only a small number of POS samples were misclassified as NEG (199 samples) and NEU (153 samples). This indicates that Bi-LSTM is relatively effective at learning semantic features associated with positive customer comments, particularly when explicit emotional expressions are present.

For the Negative (NEG) class, the model also achieved relatively good performance, correctly classifying 1,051 samples. However, 155 samples were misclassified as Neutral and 128 as Positive. This suggests that some negative comments may contain neutral expressions or weak emotional signals, making them more difficult to distinguish.

The Neutral (NEU) class exhibited the weakest performance. Only 200 samples were correctly classified, while 352 samples were misclassified as Negative and 387 as Positive. This result indicates that the model struggles to identify neutral comments accurately. One possible reason is that neutral comments often lack explicit emotional keywords and share linguistic characteristics with both positive and negative comments.

Overall, most classification errors are concentrated in the Neutral class, whereas the Negative and Positive classes are recognized more effectively. These findings highlight the limitations of the Bi-LSTM architecture in capturing complex contextual information and subtle emotional nuances in Vietnamese text. Although the model achieves reasonably good overall performance and performs well on comments with clear emotional polarity, recognizing neutral and implicitly expressed emotions remains a significant challenge that affects its reliability in real-world applications.

b, CNN

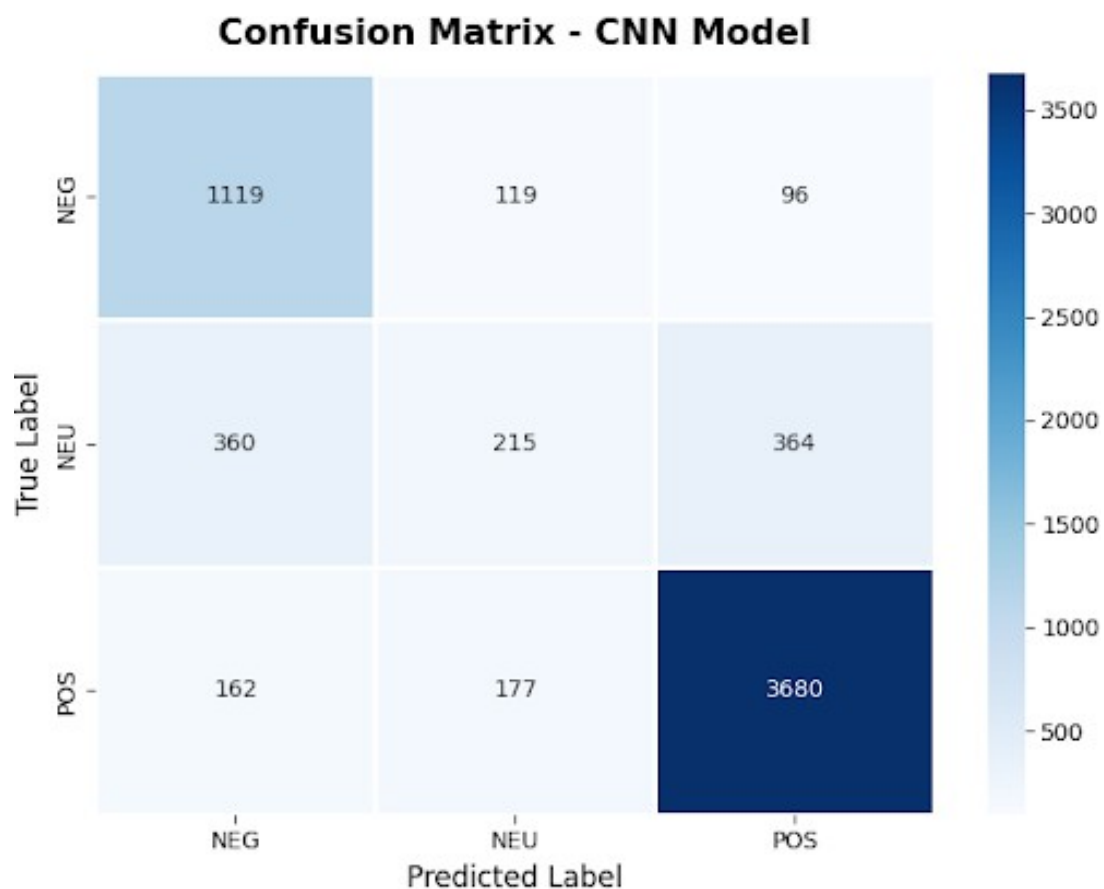


Figure 8: Confusion matrix of the CNN model .

Figure 8 presents the confusion matrix of the CNN model on the test dataset. Most values are concentrated along the main diagonal, indicating that the model correctly classified a large proportion of samples across all three emotion categories. This observation is consistent with the quantitative evaluation results, where CNN achieved an accuracy of 79.69%, making it one of the best-performing models in this study.

The Positive (POS) class achieved the highest classification performance, with 3,680 samples correctly predicted. Only a small number of positive samples were misclassified into the Negative (NEG) and Neutral (NEU) classes. This result demonstrates the model's ability to effectively learn discriminative features associated with positive customer comments, thereby

contributing significantly to the overall classification accuracy.

For the Negative (NEG) class, CNN also demonstrated strong classification capability, correctly identifying 1,119 samples. The number of samples misclassified as Neutral or Positive remained relatively low, suggesting that the convolutional layers successfully captured important local semantic patterns associated with negative emotions.

In contrast, the Neutral (NEU) class remained the most challenging category. Only 215 neutral samples were correctly classified, while a considerable number were incorrectly assigned to the Positive or Negative classes. This finding indicates that the distinction between neutral and emotional comments is less explicit in real-world datasets. Many neutral comments contain weak emotional signals or ambiguous expressions, making accurate classification more difficult.

Compared with the RNN (Bi-LSTM) model, CNN produced fewer classification errors in both the Positive and Negative classes and achieved a greater number of correct predictions along the main diagonal of the confusion matrix. This suggests that CNN is more effective at extracting local contextual features from text, particularly emotional keywords and highly expressive phrases.

Overall, the confusion matrix demonstrates that CNN performs consistently well in the task of Vietnamese customer emotion classification. Although the model still encounters difficulties in recognizing Neutral comments, it exhibits strong predictive capability for comments with clear emotional polarity. Combined with its relatively low computational cost and high inference speed, CNN provides an effective balance between classification performance and computational efficiency, making it a practical choice for real-world sentiment analysis applications.

c, Teacher PhoBERT

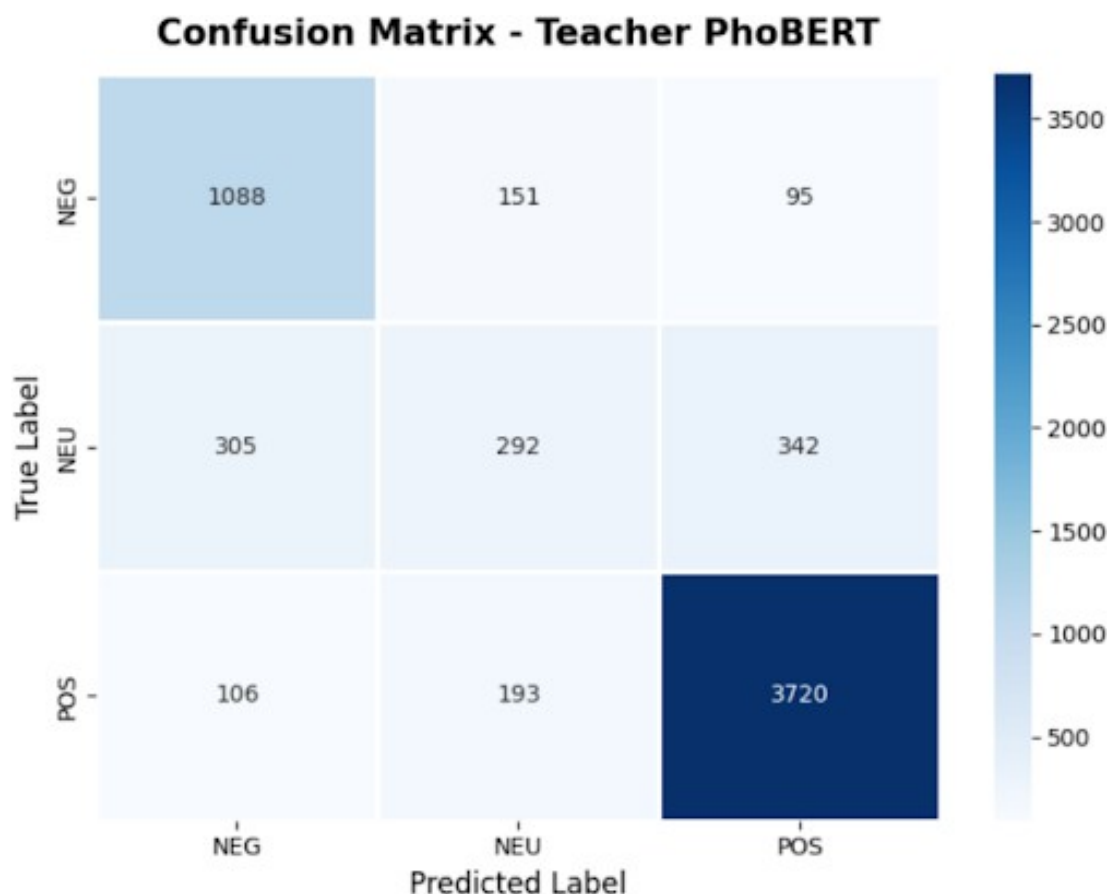


Figure 9: Confusion matrix of the Teacher Phobert model .

Figure 9 presents the confusion matrix of the Teacher PhoBERT model on the test dataset. The majority of correctly classified samples are concentrated along the main diagonal, indicating the model's strong capability in learning and representing semantic information in Vietnamese text. This observation is consistent with the quantitative evaluation results, where Teacher PhoBERT achieved the highest overall performance among all evaluated models, with an Accuracy of 81.06

For the Positive (POS) class, the model achieved the highest classification performance, correctly predicting 3,720 samples. Only 106 samples were misclassified as Negative and 193 as Neutral. These results demonstrate PhoBERT's ability to effectively recognize positive comments, even when emotional expressions are implicit or conveyed through diverse linguistic patterns.

For the Negative (NEG) class, the model correctly classified 1,088 samples. Although a small number of samples were incorrectly assigned to the Neutral (151 samples) and Positive (95 samples) classes, the overall error rate remained relatively low. This finding indicates that PhoBERT successfully captures contextual and semantic features associated with negative emotions, thereby reducing classification errors.

The Neutral (NEU) class remained the most challenging category. The model correctly identified 292 samples, while 305 samples were misclassified as Negative and 342 samples

as Positive. Nevertheless, compared with the CNN and RNN (Bi-LSTM) models, Teacher PhoBERT achieved a substantially higher number of correct predictions for the Neutral class. This improvement suggests that the contextual understanding provided by the Transformer architecture enables the model to better interpret comments containing ambiguous or weak emotional signals.

A notable observation is the relatively low level of confusion between the Positive and Negative classes. Most classification errors were concentrated in the Neutral category, indicating that the model learned clear decision boundaries between the two opposing emotional classes. This characteristic is particularly valuable in customer sentiment analysis applications, where distinguishing positive and negative feedback accurately is often of primary importance.

Overall, the confusion matrix demonstrates that Teacher PhoBERT is the most robust and comprehensive emotion classification model evaluated in this study. The model not only achieved the highest number of correct predictions for the Positive and Negative classes but also significantly improved the recognition of Neutral comments compared with the other approaches. These findings further confirm the advantages of Transformer-based architectures in capturing contextual and semantic information in Vietnamese text, thereby enhancing the effectiveness of customer emotion detection systems.[1]

3.3.2. Training results of Machine Learning algorithms

a, Naive Bayes

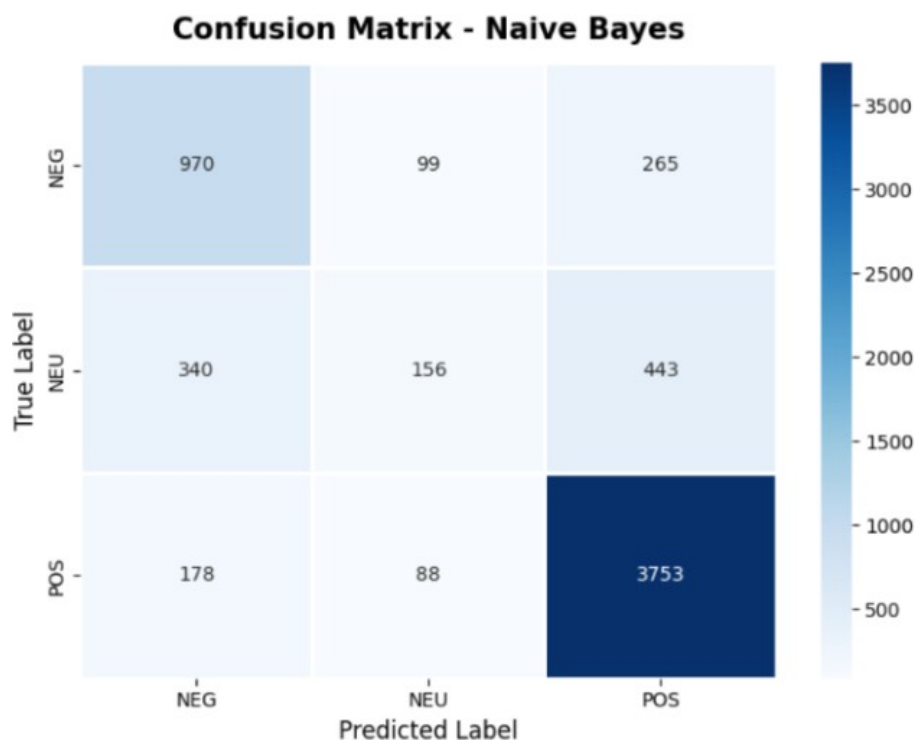


Figure 10: Presents the confusion matrix of the Naive Bayes model.

Figure 10 presents the confusion matrix of the Naive Bayes model on the test dataset. The values along the main diagonal represent correctly classified samples, while the off-diagonal values indicate classification errors. Visual inspection reveals that the model performs best on the Positive (POS) class but encounters considerable difficulty when classifying Neutral (NEU) comments.

The Positive (POS) class achieved the highest classification performance, with 3,753 samples correctly identified. Only 178 samples were misclassified as Negative and 88 samples as Neutral. This result suggests that positive comments contain distinctive lexical patterns that can be effectively captured by the probabilistic assumptions of the Naive Bayes classifier.

In contrast, the Neutral (NEU) class exhibited the lowest classification performance. Out of all neutral samples, only 156 were correctly classified, while 340 samples were incorrectly assigned to the Negative class and 443 samples to the Positive class. These findings indicate that Naive Bayes struggles to distinguish neutral comments from emotionally polarized comments. One possible explanation is that the model relies primarily on word occurrence frequencies and cannot effectively capture contextual relationships within a sentence.

For the Negative (NEG) class, the model correctly classified 970 samples. However, 265 samples were misclassified as Positive and 99 samples as Neutral. This result reflects a degree of overlap between the linguistic characteristics of negative and positive comments, making it difficult for the model to establish clear decision boundaries between the two emotion categories.

A notable observation from the confusion matrix is that the majority of classification errors are concentrated in the Neutral class. Rather than being recognized as a distinct category, many neutral comments are incorrectly assigned to either the Positive or Negative classes. This phenomenon is commonly observed in traditional probabilistic classifiers when processing text data containing subtle semantic distinctions and implicit emotional expressions.

Overall, the confusion matrix demonstrates that Naive Bayes is capable of effectively recognizing comments with explicit emotional polarity, particularly positive comments. However, its performance decreases substantially when dealing with neutral or weakly expressed emotions. These findings suggest that Naive Bayes serves as a useful baseline model due to its simplicity and computational efficiency, but it remains less effective than more advanced architectures such as Bi-LSTM, CNN, and Teacher PhoBERT for Vietnamese customer emotion classification.

b, Logistic Regression



Figure 11: Illustrates the confusion matrix of the Logistic Regression model.

Figure 11 presents the confusion matrix of the Logistic Regression model on the test dataset. The majority of correctly classified samples are concentrated along the main diagonal, indicating that the model successfully learned important discriminative features from the text data. Although misclassifications still occur, particularly for the Neutral (NEU) class, Logistic Regression demonstrates stronger classification capability than traditional probabilistic approaches in distinguishing between positive and negative customer emotions.

The Positive (POS) class achieved the highest classification performance, with 3,773 samples correctly identified. Only 136 samples were misclassified as Negative and 110 samples as Neutral. These results indicate that Logistic Regression effectively captures linguistic patterns associated with positive emotions, resulting in a relatively low misclassification rate.

For the Negative (NEG) class, the model correctly classified 1,045 samples. Meanwhile, 118 samples were incorrectly assigned to the Neutral class and 171 samples to the Positive class. Compared with Naive Bayes, the number of correctly classified negative samples increased, suggesting that Logistic Regression establishes a more effective decision boundary between emotion categories.

Similar to the other evaluated models, the Neutral (NEU) class remained the most challenging category. The model correctly identified only 168 samples, while 310 samples were misclassified as Negative and 461 samples as Positive. This result suggests that neutral comments often lack distinctive emotional indicators, making it difficult for the model to

accurately determine the appropriate emotion label.

A notable observation from the confusion matrix is that the number of direct misclassifications between the Positive and Negative classes is relatively low compared with errors involving the Neutral class. This indicates that Logistic Regression learned clear boundaries between the two opposing emotional categories, whereas neutral comments are still frequently assigned to one of the emotional extremes.

Overall, the confusion matrix demonstrates that Logistic Regression delivers stable classification performance, particularly for the Positive and Negative classes. However, its ability to recognize Neutral comments remains limited, highlighting the difficulty of handling ambiguous or weakly expressed emotions using linear classification methods. These findings confirm that Logistic Regression serves as an effective baseline model for Vietnamese customer emotion classification, although a noticeable performance gap remains compared with deep learning and Transformer-based approaches that can better capture contextual and semantic information.

c, Random Forest

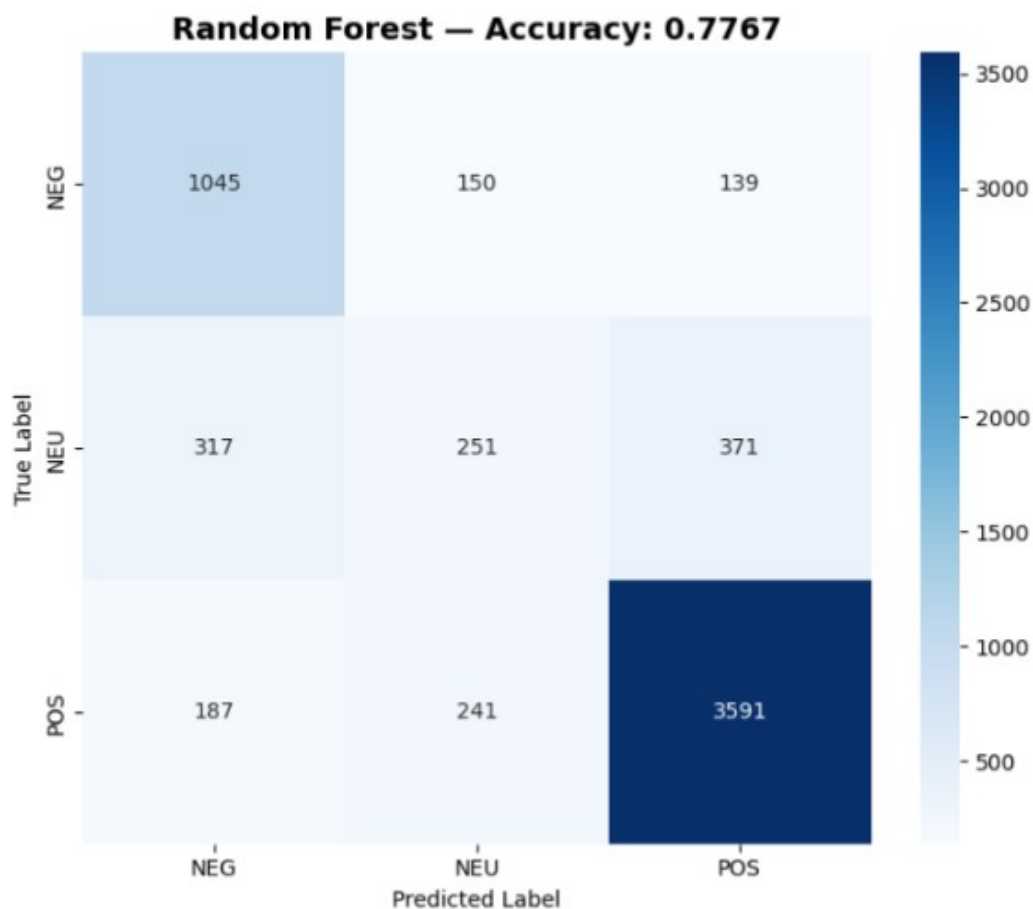


Figure 12: Illustrates the confusion matrix of the Random Forest model.

Figure 12 presents the confusion matrix of the Random Forest model on the test dataset. The distribution of values along the main diagonal indicates that the model correctly classified a large proportion of samples across all emotion categories. This observation is consistent with the quantitative evaluation results, where Random Forest achieved an accuracy of 77.67%. Although the model demonstrates relatively strong performance in recognizing Positive and Negative comments, the Neutral class remains the most challenging category.

The Positive (POS) class achieved the highest classification performance, with 3,591 samples correctly identified. In comparison, 187 samples were misclassified as Negative and 241 samples as Neutral. While the number of correct predictions remained high, the increase in misclassification errors compared with Logistic Regression suggests that Random Forest was unable to fully exploit all semantic characteristics present in the text data.

For the Negative (NEG) class, the model correctly classified 1,045 samples. Meanwhile, 150 samples were incorrectly assigned to the Neutral class and 139 samples to the Positive class. These results indicate that Random Forest maintains a relatively strong capability in recognizing negative comments while keeping the level of confusion between the two opposing emotion categories at a moderate level.

The Neutral (NEU) class achieved better performance than several traditional machine learning models, with 251 correctly classified samples. However, 317 samples were misclassified as Negative and 371 samples as Positive. This finding suggests that although Random Forest improves the recognition of neutral comments compared with simpler classifiers, distinguishing comments that contain weak or ambiguous emotional signals remains difficult.

A notable observation from the confusion matrix is that the majority of classification errors are concentrated in the Neutral class. In contrast, direct confusion between the Positive and Negative classes is relatively limited. This indicates that Random Forest successfully learns discriminative features that separate the two opposing emotional categories but still struggles to establish clear decision boundaries for neutral comments.

Overall, the confusion matrix demonstrates that Random Forest provides stable performance for the Vietnamese customer emotion classification task. The model achieves a high number of correct predictions for the Positive and Negative classes and shows noticeable improvement in recognizing Neutral comments compared with Naive Bayes and Logistic Regression. Nevertheless, the results suggest that there is still room for improvement through advanced deep learning and Transformer-based architectures, which are better equipped to capture contextual and semantic relationships within text data.

=> Conclusion: The confusion matrices show significant differences between the models in identifying emotions from Vietnamese comments. Although all models identify both classes relatively well. Positive and Negative, class Neutral This remains the biggest challenge due to the blurred and often contextually sensitive nature of emotional boundaries.

Among the models evaluated, Teacher PhoBERT The most stable and accurate classification ability is demonstrated when the highest number of correct predictions are concentrated along the main diagonal. CNN It also achieved very promising results, with the ability to identify

positive and negative comments almost equivalent to PhoBERT but with a simpler architecture. Conversely, RNN (Bi-LSTM) Many instances of confusion still occur in the Neutral class, indicating limitations in fully exploiting the semantics of the text.

For traditional machine learning models, Logistic Regression and Random Forest yielding relatively stable results, while Naive Bayes They face more difficulties when processing comments with complex emotional nuances. In general, models with better contextual learning capabilities tend to produce fewer classification errors and provide clearer boundaries between emotion classes. This confirms the advantage of deep learning methods, especially PhoBERT and CNN, in the problem of detecting customer emotions from Vietnamese comments.

3.3.3. Streamlit Interface Visualization

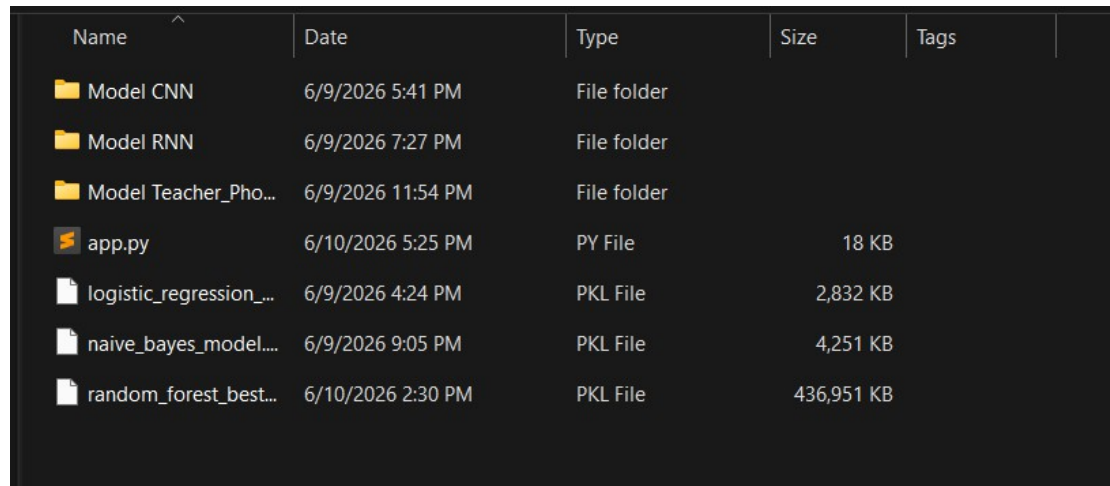
The system is implemented as a web interface using Streamlit, allowing users to input comments in Vietnamese and predict customer sentiment using three labels: Positive, Negative, and Neutral. The interface is designed in a dark style with a clear layout, including a configuration bar on the left and a model testing area in the center.

On the left sidebar, users can select prediction modes, including choosing a single model or comparing all six models. The models integrated into the system include Logistic Regression, Naive Bayes, Random Forest, RNN, CNN, and Teacher Phobert. Additionally, the interface displays an overview of the problem, the number of classification classes, and the output sentiment labels.

The central part of the interface includes the system title, dataset information, a list of models used, and a comment input field for testing. After the user enters a comment and clicks the Predict button, the system processes the input text, applies the selected model or all six models, and then displays the prediction results in a table.

How to run Streamlit using Anaconda Prompt:

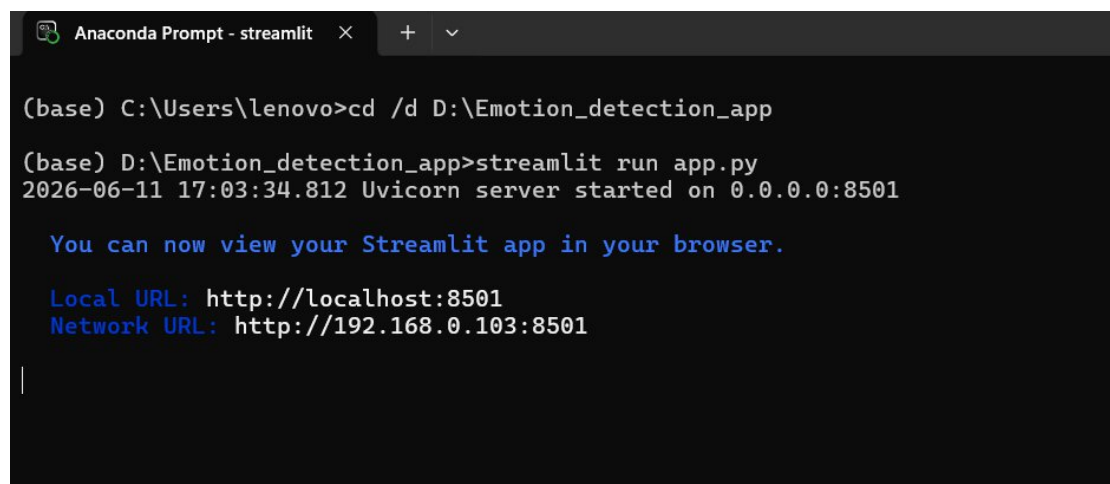
Step 1: To run the Streamlit interface, you first need to place the app.py file in the same directory as the models saved after training. The directory structure can be organized as follows:



Name	Date	Type	Size	Tags
Model CNN	6/9/2026 5:41 PM	File folder		
Model RNN	6/9/2026 7:27 PM	File folder		
Model Teacher_Ph...	6/9/2026 11:54 PM	File folder		
app.py	6/10/2026 5:25 PM	PY File	18 KB	
logistic_regression_...	6/9/2026 4:24 PM	PKL File	2,832 KB	
naive_bayes_model....	6/9/2026 9:05 PM	PKL File	4,251 KB	
random_forest_best...	6/10/2026 2:30 PM	PKL File	436,951 KB	

Figure 13: The file containing the model after training and app.py

Step 2: Open Anaconda Prompt, navigate to the folder containing the app.py file, and run the application:



```
Anaconda Prompt - streamlit X + v
(base) C:\Users\lenovo>cd /d D:\Emotion_detection_app
(base) D:\Emotion_detection_app>streamlit run app.py
2026-06-11 17:03:34.812 Uvicorn server started on 0.0.0.0:8501

You can now view your Streamlit app in your browser.

Local URL: http://localhost:8501
Network URL: http://192.168.0.103:8501
```

Figure 14: Run the Streamlit application on Anaconda Prompt.

Step 3: After running, Streamlit will open the interface in your browser at the address above:

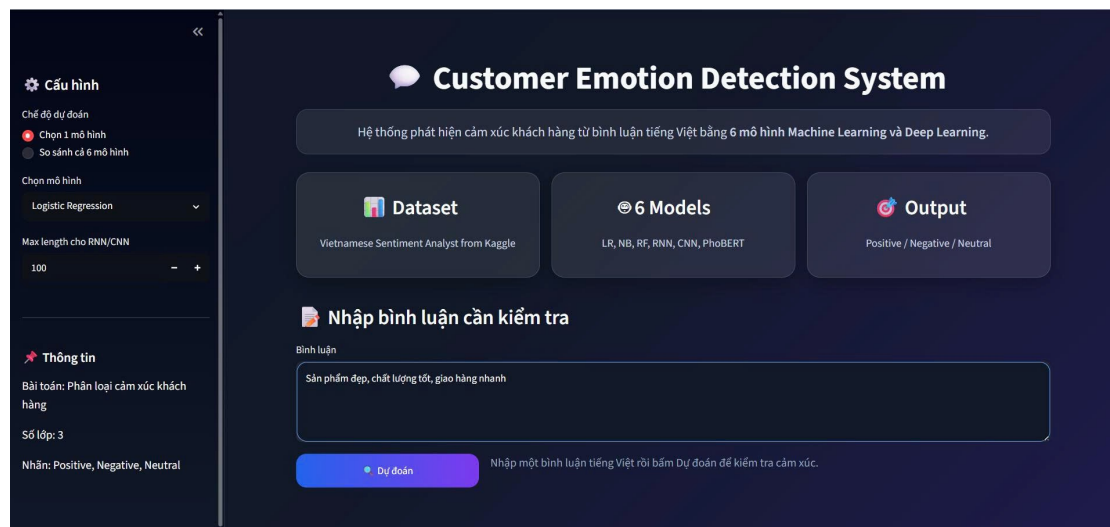


Figure 15: The main interface of the Customer Emotion Detection System.

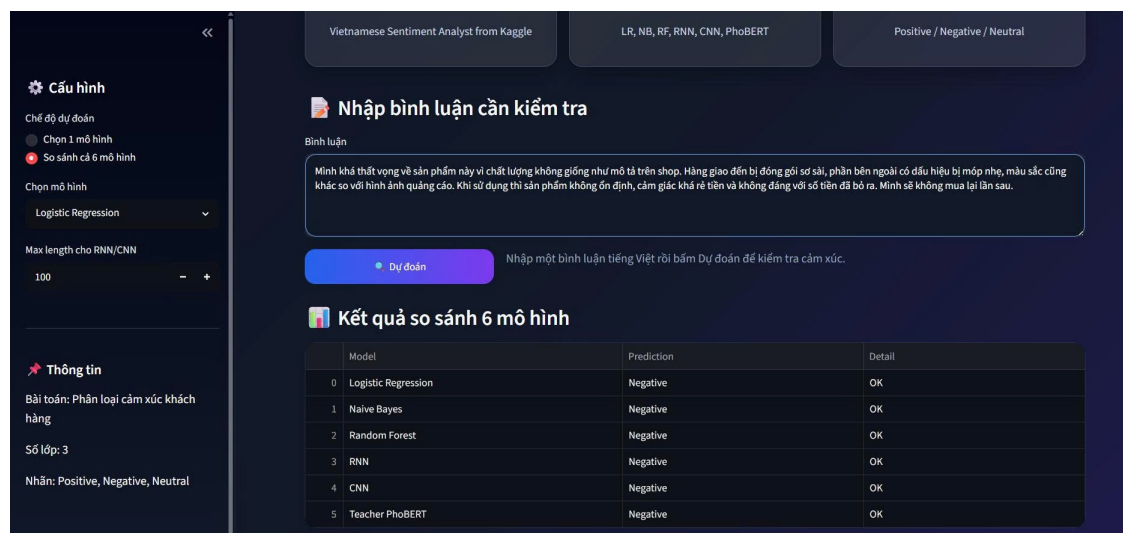


Figure 16: Predicting negative comments using 6 models.

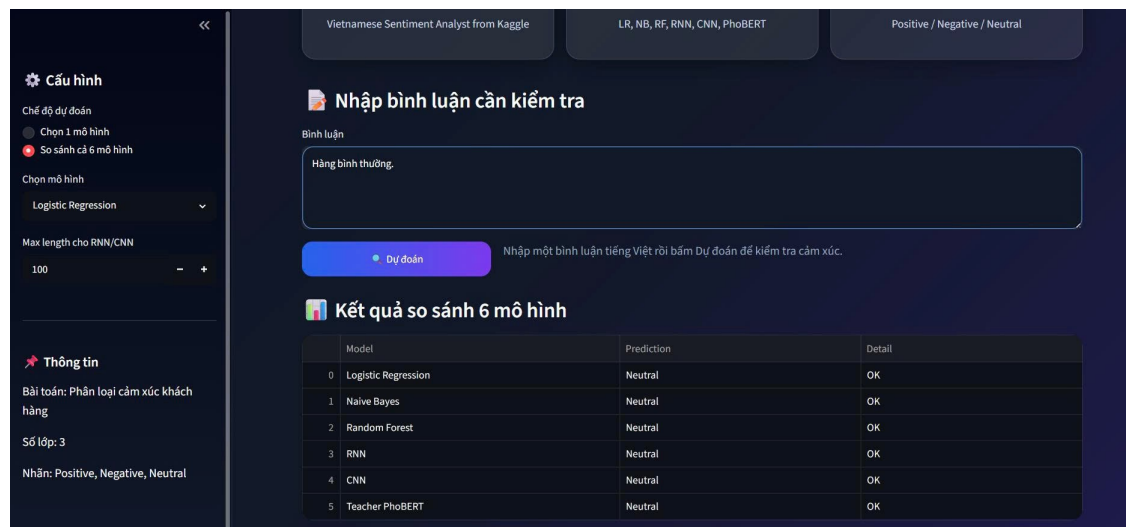


Figure 17: Predicting neutral comments using 6 models.

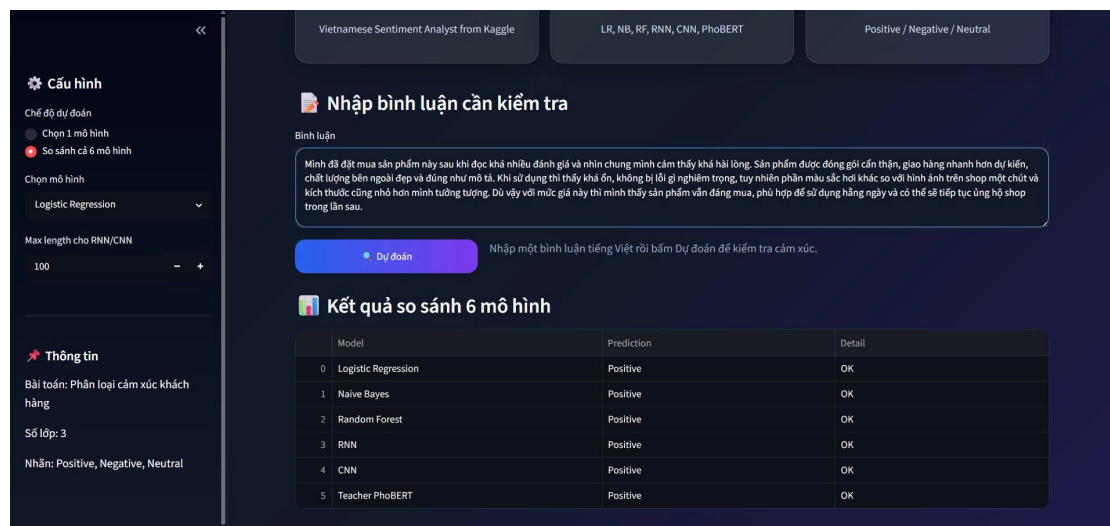


Figure 18: The results predict positive comments using 6 models.

3.4. Error Analysis

This section analyzes the differences in predictions across the six models on two real comments from the demo interface, in order to clarify the causes of disagreement among models when handling comments with ambiguous sentiment.

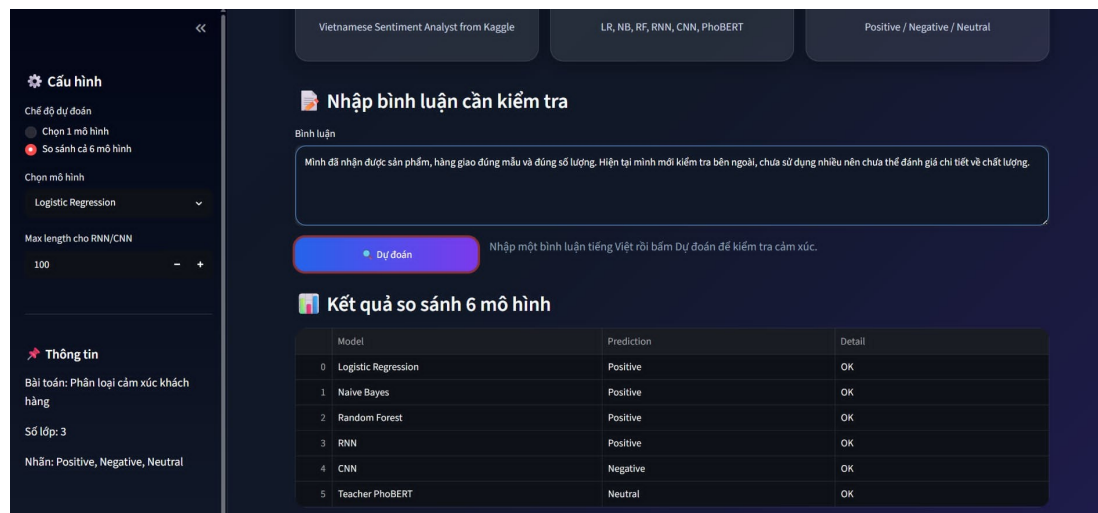


Figure 19: Result analysis image failed.

This comment is a conditionally neutral statement: the first clause confirms correct delivery (positive), but the second clause shows the customer deliberately withholding a quality judgment the most appropriate label is Neutral. LR, NB, RF, and RNN all predicted Positive, likely because they are dominated by a high-TF-IDF-weight phrase confirming correct delivery, while the reserved clause that follows is overshadowed. CNN predicted Negative because its n-gram kernels picked up phrases containing a negation word that often co-occurs with complaints, causing global max pooling to select a misleading negative local feature. Only PhoBERT predicted Neutral the closest to the correct label thanks to self-attention balancing the semantics of the two contrasting clauses within the sentence.

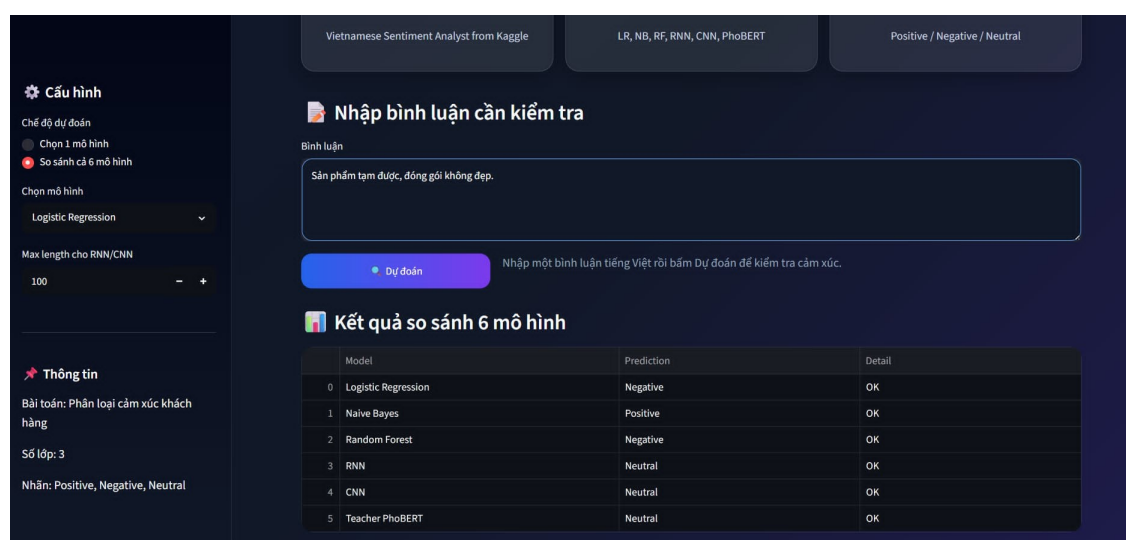


Figure 20: Result analysis image failed

This is a mixed-sentiment sentence: the first clause is mildly neutral-to-positive, while the second clause is locally negative the most appropriate label is Neutral. LR and RF predicted Negative because the negative phrase carries a strong NEGATIVE TF-IDF weight that dominates

the entire short sentence. NB predicted Positive instead, due to ComplementNB's mechanism of estimating probabilities from complement classes, which skewed the first phrase toward the Positive class. The three deep learning models (RNN, CNN, PhoBERT) all agreed on Neutral, demonstrating a much better ability to integrate semantics across multiple clauses compared to the discrete word-counting features of TF-IDF.

Classical machine learning models (especially LR and RF) tend to be dominated by clearly polarized words or phrases regardless of overall context, while ComplementNB can sometimes produce hard-to-interpret results due to its complement-based probability mechanism. CNN, despite its multi-scale design, can still select a misleading negative local feature through Max Pooling. PhoBERT showed the highest stability and accuracy in both cases, reinforcing its role as the teacher model. Overall, mixed-sentiment or conditionally neutral comments are the main source of disagreement among the models, and should be given greater representation in the training data in future work.

4. Discussion

4.1. Analysis of the Systems Advantages

The experimental results demonstrate that the proposed sentiment analysis system is capable of effectively classifying Vietnamese customer comments into three emotional categories: Positive, Negative, and Neutral. One of the most significant advantages of the system lies in its comprehensive evaluation framework, which combines both traditional Machine Learning techniques and modern Deep Learning approaches. By evaluating six different models under the same experimental conditions, the study provides a clear and objective comparison regarding classification performance, computational cost, and deployment feasibility.

Another important advantage is the use of a unified preprocessing pipeline. All models were trained on the same cleaned and normalized dataset, ensuring fairness in evaluation and reducing the influence of external factors. This design allows the observed performance differences to primarily reflect the capabilities of the models themselves rather than inconsistencies in data preparation. The experimental results indicate that Teacher PhoBERT achieved the highest classification performance among all evaluated models. As a Transformer-based language model specifically pre-trained for Vietnamese, PhoBERT demonstrates superior capability in capturing semantic relationships, contextual information, and complex linguistic structures. This advantage becomes particularly evident when processing comments containing indirect expressions, mixed sentiments, or ambiguous wording. The model is able to understand contextual dependencies that traditional approaches often fail to capture.

CNN also demonstrated notable strengths. While its classification performance was slightly lower than PhoBERT, it achieved significantly lower computational requirements and faster inference speed. This balance between accuracy and efficiency makes CNN an attractive solution for practical applications where computational resources are limited.

Furthermore, the study confirms that traditional machine learning models remain competitive baselines. Logistic Regression, Naive Bayes, and Random Forest achieved relatively strong results despite their simpler architectures. These models require substantially fewer computational resources and can be deployed easily in environments where GPU acceleration is unavailable. Therefore, the system offers flexibility by providing multiple deployment options depending on application requirements and hardware constraints. Overall, the proposed system successfully combines predictive performance, computational efficiency, and practical applicability, making it suitable for real-world sentiment analysis tasks involving Vietnamese customer feedback.

4.2. Current Limitations of the Study

Despite achieving promising results, the study still faces several limitations that should be acknowledged.

First, the dataset used in this research contains a degree of class imbalance. Positive comments account for a significantly larger proportion of the dataset compared to Neutral and Negative comments. Although weighted loss functions and additional evaluation metrics such as Precision, Recall, and F1-score were employed to mitigate this issue, the imbalance may still influence the learning process and lead to biased predictions toward the majority class.

Second, the system focuses exclusively on three sentiment categories. Human emotions are inherently complex and often cannot be accurately represented using only Positive, Negative, and Neutral labels. In many real-world scenarios, customer comments may simultaneously contain multiple emotional states such as satisfaction, disappointment, frustration, excitement, or uncertainty. The current classification framework cannot fully capture such fine-grained emotional variations.

Another limitation is related to language complexity. Vietnamese customer comments frequently include slang, abbreviations, regional expressions, spelling variations, emojis, and informal writing styles. Although the deep learning models demonstrate robustness against such linguistic noise, misclassification still occurs when comments are highly ambiguous or contain sarcasm, irony, and implicit emotional expressions.

From a technical perspective, Teacher PhoBERT requires substantial computational resources for both training and deployment. Its large number of parameters and storage requirements may limit practical implementation on edge devices or low-resource environments. While the model achieves the best classification performance, the associated hardware costs may not always be feasible for small organizations or real-time systems.

Finally, the study only evaluates models on a single Vietnamese sentiment dataset. Although the results are encouraging, additional experiments on datasets collected from different domains such as social media, e-commerce platforms, tourism services, healthcare systems, or educational environments would provide stronger evidence regarding model generalization and robustness.

4.3. Evaluating the Effectiveness of Improvement Solutions

Several improvement strategies were incorporated into the system design to address common challenges in sentiment analysis, and the experimental results demonstrate their effectiveness. One of the most impactful improvements was the adoption of advanced text preprocessing techniques. Data cleaning, normalization, label encoding, and feature transformation helped reduce noise and improve the quality of the training data. This preprocessing pipeline contributed significantly to the stability and consistency of model performance.

The use of Dual TF-IDF representations for machine learning models also proved beneficial. By combining Word TF-IDF and Character TF-IDF features, the models became more capable of handling misspellings, abbreviations, and informal writing styles commonly found in online customer comments. As a result, traditional machine learning methods achieved stronger performance than expected and remained competitive with some deep learning approaches. For deep learning models, architectural improvements played an important role. The Bi-LSTM architecture enabled the model to capture contextual information from both directions of a sentence, while the multi-kernel TextCNN architecture successfully extracted sentiment patterns at different phrase lengths. These design choices contributed to improved classification performance compared with simpler neural network architectures.

The most effective improvement was the utilization of Teacher PhoBERT with weighted cross-entropy loss. By leveraging knowledge acquired during large-scale Vietnamese language pre-training, PhoBERT demonstrated superior contextual understanding and achieved the highest overall performance across all evaluation metrics. Additionally, the use of class weighting helped alleviate the impact of dataset imbalance and improved the recognition of minority sentiment classes.

Taken together, these improvement strategies significantly enhanced both predictive accuracy and system robustness. The results confirm that combining appropriate preprocessing techniques, feature engineering, model optimization, and language-specific pre-trained models can substantially improve sentiment classification performance for Vietnamese text.

4.4. Challenges Encountered During Implementation During the implementation process, several challenges were encountered.

One major challenge involved data quality. Customer comments collected from online platforms often contain incomplete sentences, spelling errors, slang expressions, repeated characters, emojis, and inconsistent formatting. Considerable effort was required to design an effective preprocessing pipeline capable of reducing noise while preserving meaningful sentiment information.

Another challenge was the computational cost associated with training deep learning models. While traditional machine learning models could be trained relatively quickly, the training process for PhoBERT required substantially longer execution times and greater memory consumption. Managing GPU resources efficiently and preventing training interruptions became

important considerations throughout the experimentation phase.

Hyperparameter optimization also presented difficulties. Different models required different configurations, and identifying suitable combinations of learning rates, batch sizes, dropout rates, and model-specific parameters involved extensive experimentation. GridSearchCV for machine learning models and repeated tuning for deep learning architectures increased overall development time. In addition, class imbalance posed a persistent challenge during model training and evaluation. Models tended to perform better on the dominant Positive class while struggling with the Neutral category. This issue required the use of weighted loss functions, balanced evaluation metrics, and careful interpretation of experimental results.

Finally, ensuring a fair comparison among six fundamentally different algorithms was itself challenging. Consistent preprocessing procedures, identical train-test splits, and unified evaluation metrics were necessary to guarantee that performance differences accurately reflected model capabilities rather than experimental inconsistencies.

5. Conclusion

5.1. Achievements

This study successfully developed and evaluated a sentiment analysis system for Vietnamese customer comments using both Machine Learning and Deep Learning approaches. Six representative models, including Naive Bayes, Logistic Regression, Random Forest, RNN, CNN, and Teacher PhoBERT, were implemented and assessed under a unified experimental framework.

The experimental results demonstrated that Teacher PhoBERT achieved the highest classification performance, reaching an accuracy of 81.06% and outperforming all other evaluated models. CNN emerged as a strong alternative by providing an excellent balance between predictive performance and computational efficiency. Traditional machine learning models also produced competitive results while requiring significantly fewer computational resources.

Beyond model evaluation, the study successfully established a complete sentiment analysis pipeline consisting of data collection, preprocessing, feature extraction, model training, performance evaluation, and prediction generation. The system effectively processes Vietnamese textual data and demonstrates the feasibility of applying modern Natural Language Processing techniques to customer feedback analysis.

The research therefore achieved its primary objective of identifying effective approaches for Vietnamese sentiment classification while providing valuable insights into the trade-offs between accuracy, efficiency, and deployment feasibility.

5.2. Practical Applications and Future Development Potential

5.2.1. *Practical Applicability*

The proposed system has considerable practical value across multiple domains. In e-commerce platforms, sentiment analysis can help businesses automatically monitor customer opinions regarding products and services, enabling faster responses to customer concerns and improved decision-making processes. In social media monitoring, the system can be used to analyze public reactions toward brands, marketing campaigns, or newly launched products. Organizations can leverage these insights to evaluate customer satisfaction and adjust business strategies accordingly.

Customer service departments may also benefit from automated sentiment analysis by prioritizing negative feedback and identifying urgent complaints requiring immediate attention. This capability can improve customer experience and reduce response times.

Furthermore, the system can support market research activities by providing large-scale analysis of consumer opinions, trends, and preferences. Because the solution is designed specifically for Vietnamese text, it addresses a practical need in the rapidly growing Vietnamese digital ecosystem.

5.2.2. *Future Development Directions*

Although the current system achieves satisfactory performance, several opportunities exist for future enhancement. One promising direction is the expansion of emotion categories beyond the current three-class framework. Fine-grained emotion recognition involving categories such as happiness, anger, sadness, surprise, and frustration would provide more detailed customer insights.

Future research may also investigate data augmentation techniques and larger datasets to improve model generalization. Collecting data from multiple domains would help increase robustness and reduce domain dependency.

Another potential improvement involves deploying lightweight transformer models through knowledge distillation or model compression techniques. Such approaches could maintain high classification performance while reducing memory consumption and inference costs, making deployment more practical on mobile and edge devices.

Additionally, multimodal sentiment analysis could be explored by combining textual information with images, audio recordings, or user interaction data. This would allow the system to capture emotional signals that may not be explicitly expressed in text alone. Finally, future work may focus on developing a real-time web-based or mobile application integrated with interactive dashboards. Such a system could provide businesses with continuous sentiment monitoring, visualization tools, and automated reporting capabilities, further enhancing the practical impact of sentiment analysis technology in real-world environments.

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